PM · SWE5006 / TEAM 14

A month of moments. One film

Rewind



Small private groups. Sealed moments. A shared reveal.

Contribute → Seal → Compile → Watch

SWE5006 · TEAM 14 · 27 OCTOBER 2026
Gopal Guruprasath

Everyday moments need a shared reason to return

Product view · private-group journey

01

Contribute

Photo or video
2–10 members

02

Seal

Private until reveal
28-day cycle

03

Watch

One group film
24-hour premiere

5 moments / 30 seconds per member each week

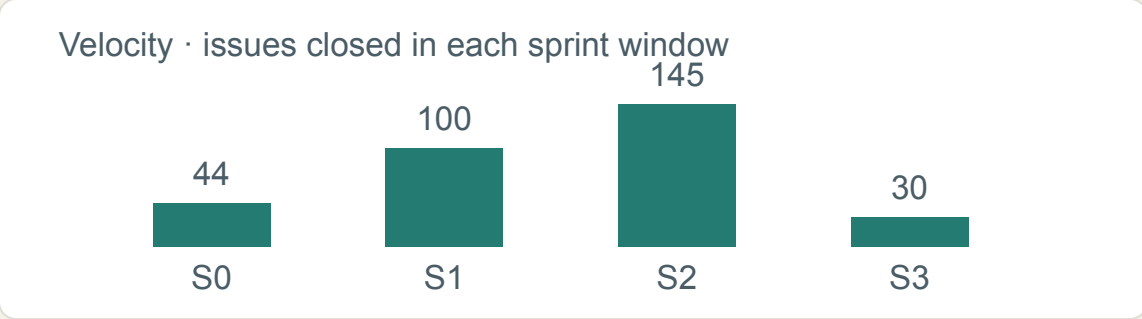
Problem. Photos disappear into busy chats; shared memories become scattered.

Scope. Accounts, invitation, capture, chat, reminders, film and Archive in a web/PWA product.

Value. A bounded contribution habit leads to one shared reveal.

Four sprints delivered the hosted private-group journey

Sprint	Milestone
0 · 1–12 Sep	Runtime foundation
1 · 13–26 Sep	Integrated capture-to-film
2 · 27 Sep–10 Oct	Real accounts · S3 · PostgreSQL · Cognito
3 · 11–24 Oct	Stability · recovery · regression tests



Team · equal effort. Nguyen Kim Long · Gopal Guruprasath · Jiang Jiayu · Phan Mai Tan Loi · Sang Haoxiang.
20% each · 10 person-days each · agreed split.

Done. Outcome checked + fast tests + green Quality + merge to dev. Batch reviews resolve blocking findings.

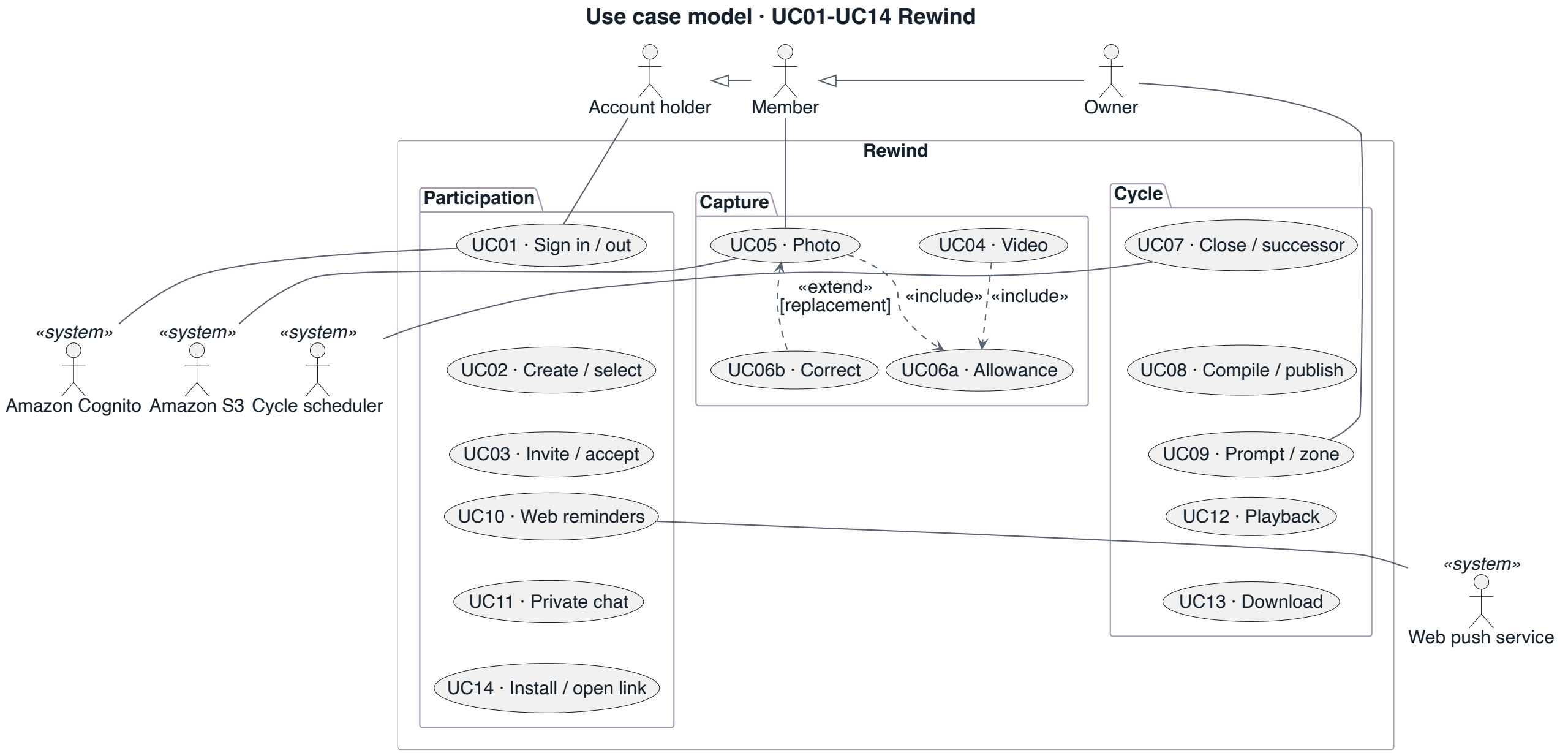
Ceremonies. Project planning; Sprint Planning and Review each sprint. Backlog → stories → sprint plans → review → retro.

Retro → next sprint. Integration gaps in S1 → real adapters in S2; stale state in S2 → interaction tests in S3.

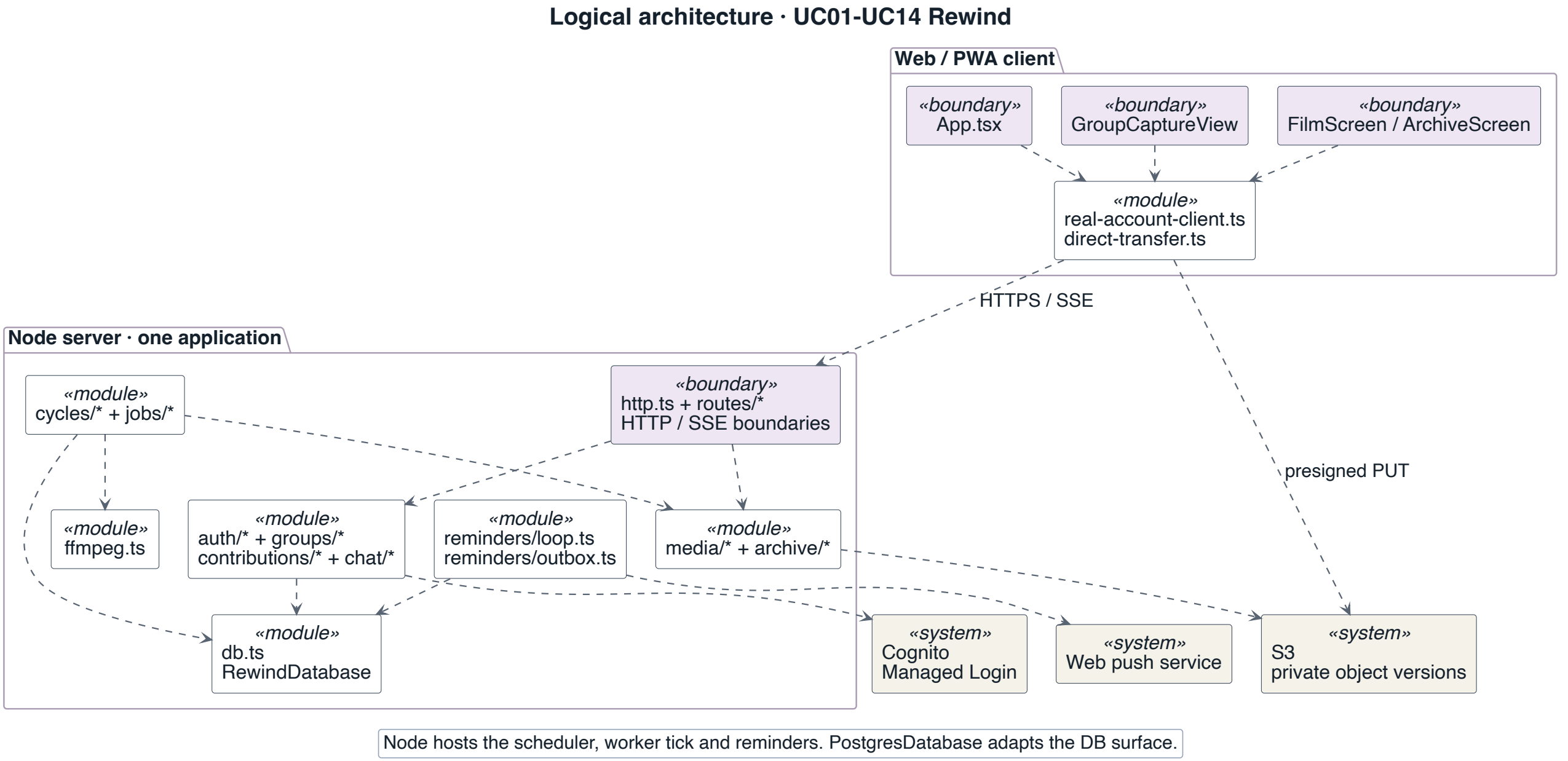
26 OCT · OPEN #587 focused runner · #588 chat labels · #527 Safari/PWA integration · #362 report assembly

Full delivery register → B10

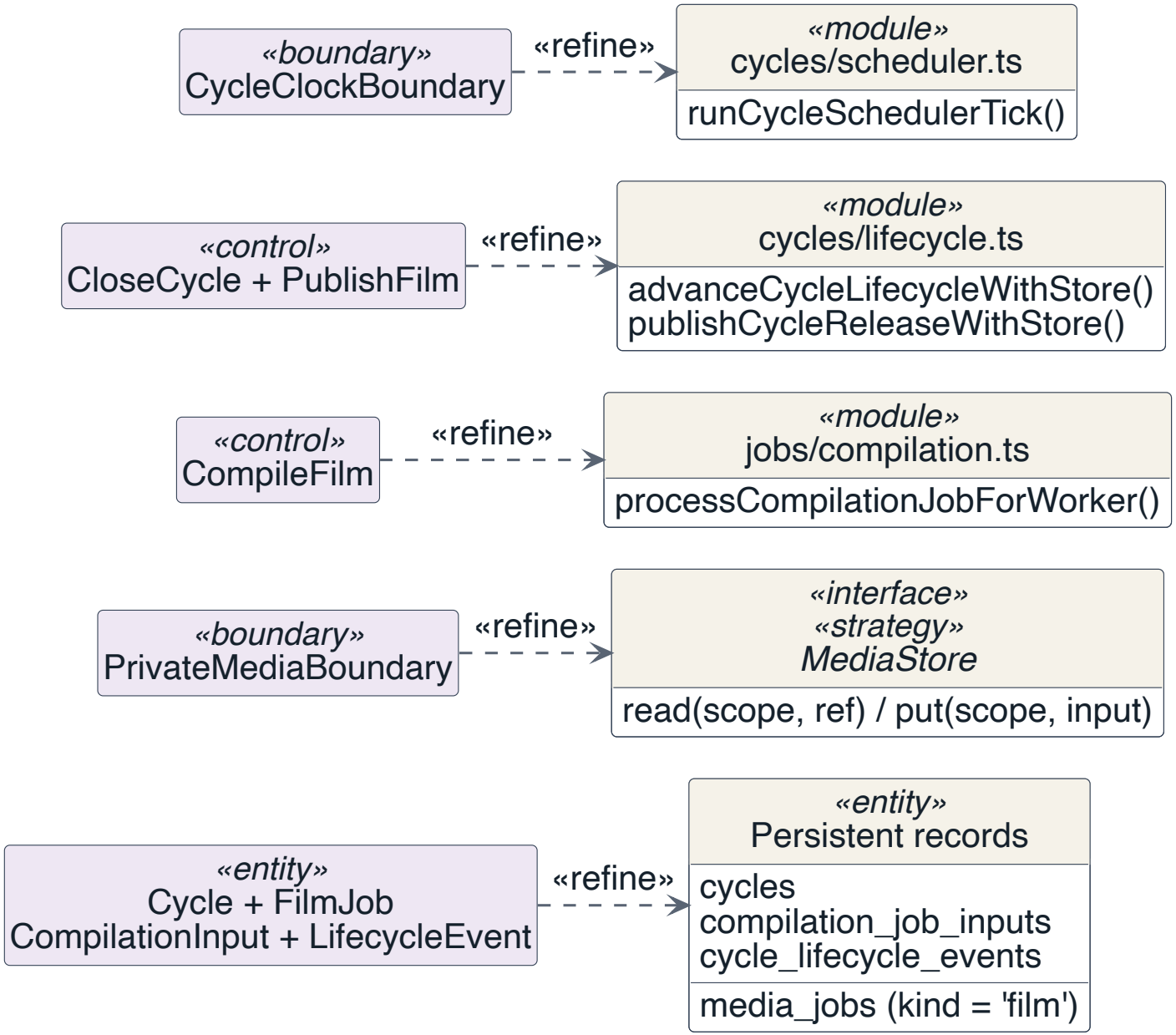
Fourteen use cases follow one private-group journey



One Node application keeps rules close to transactions

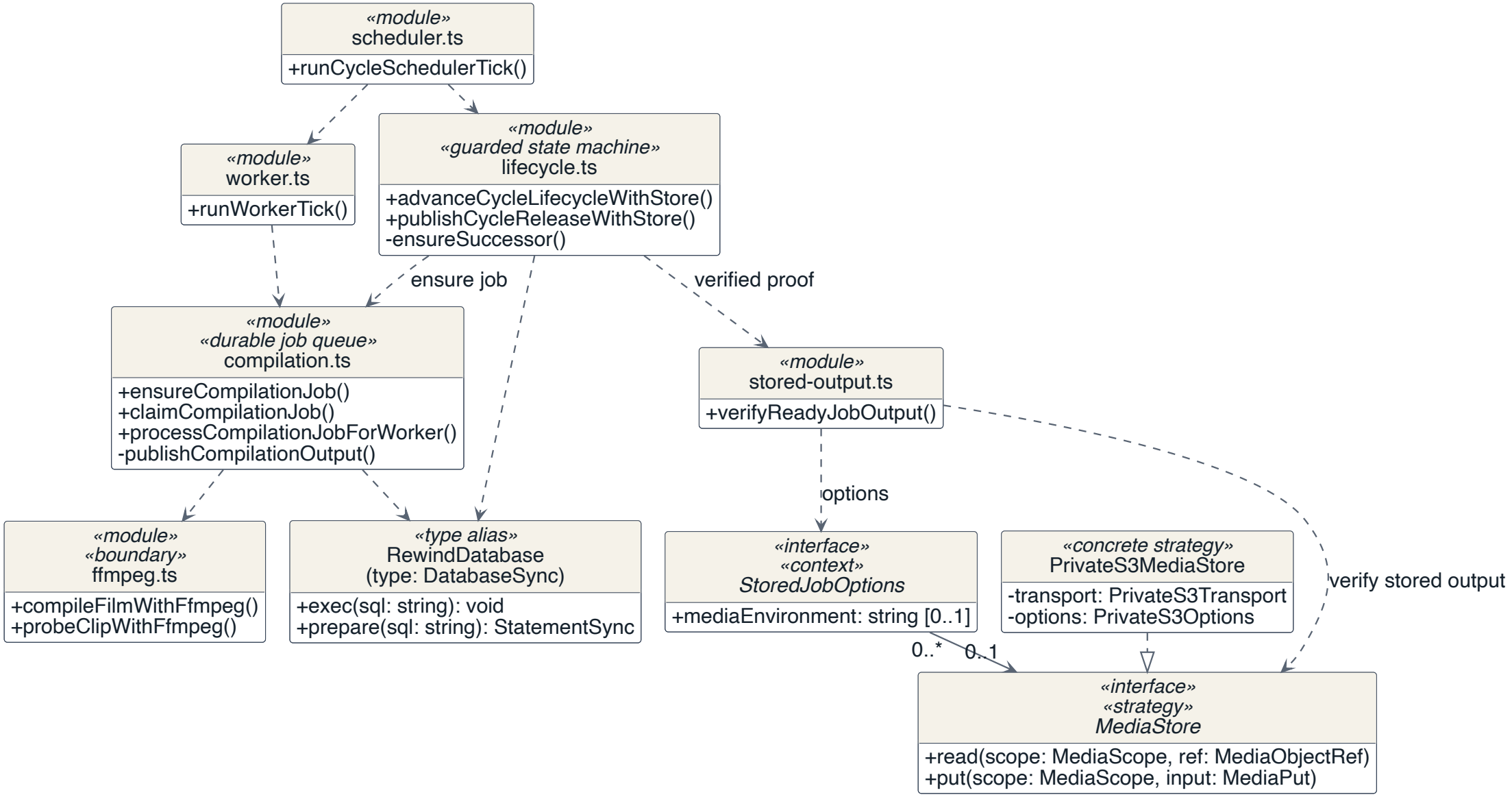


Analysis to design · UC07-UC08 Close cycle and release film



Durable claims keep repeated cycle work safe

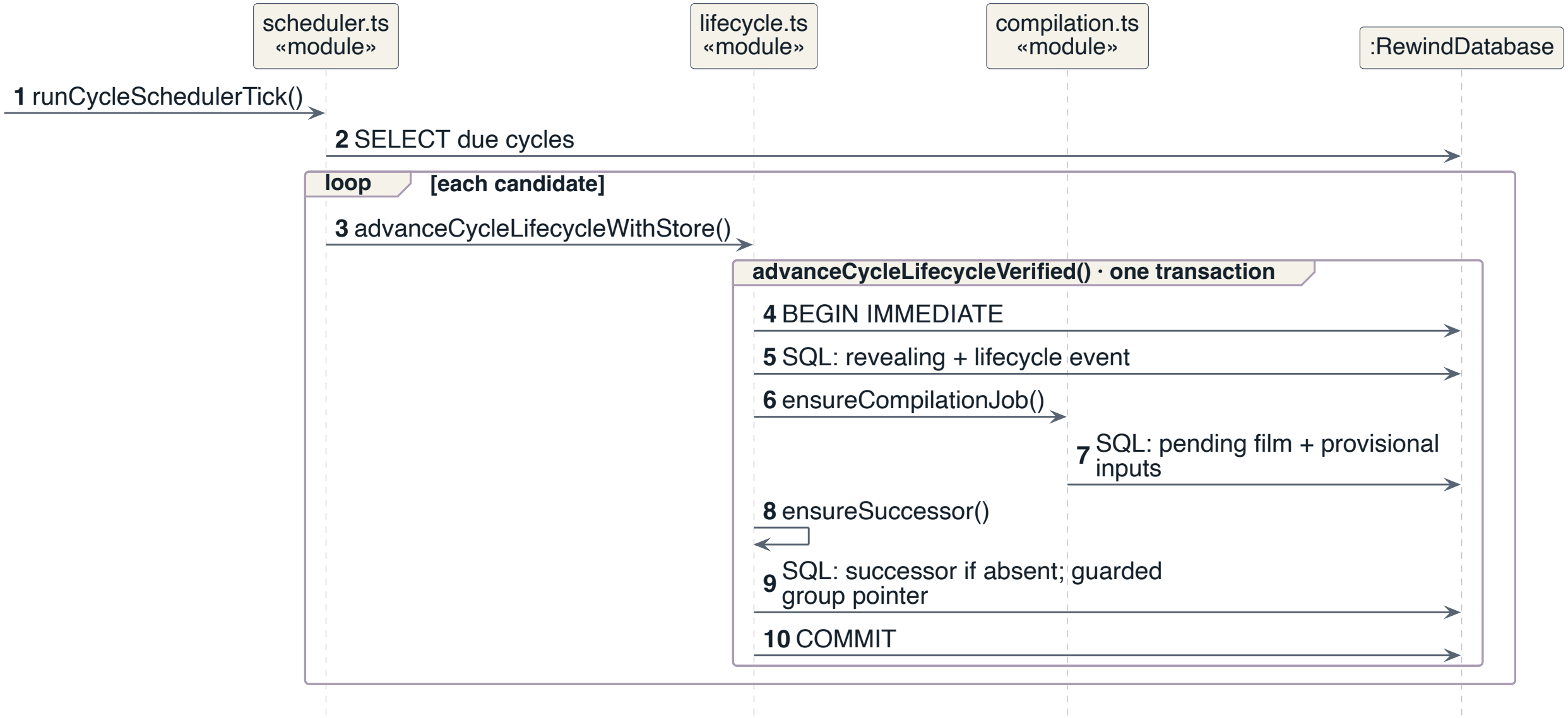
Design classes · UC07-UC08 Close cycle and release film



State machine = persisted states + guards. Job queue = records + generation fence.
Functions belong to «module» elements. Storage interface is the Strategy seam.

A repeated tick reuses work and fences stale completion

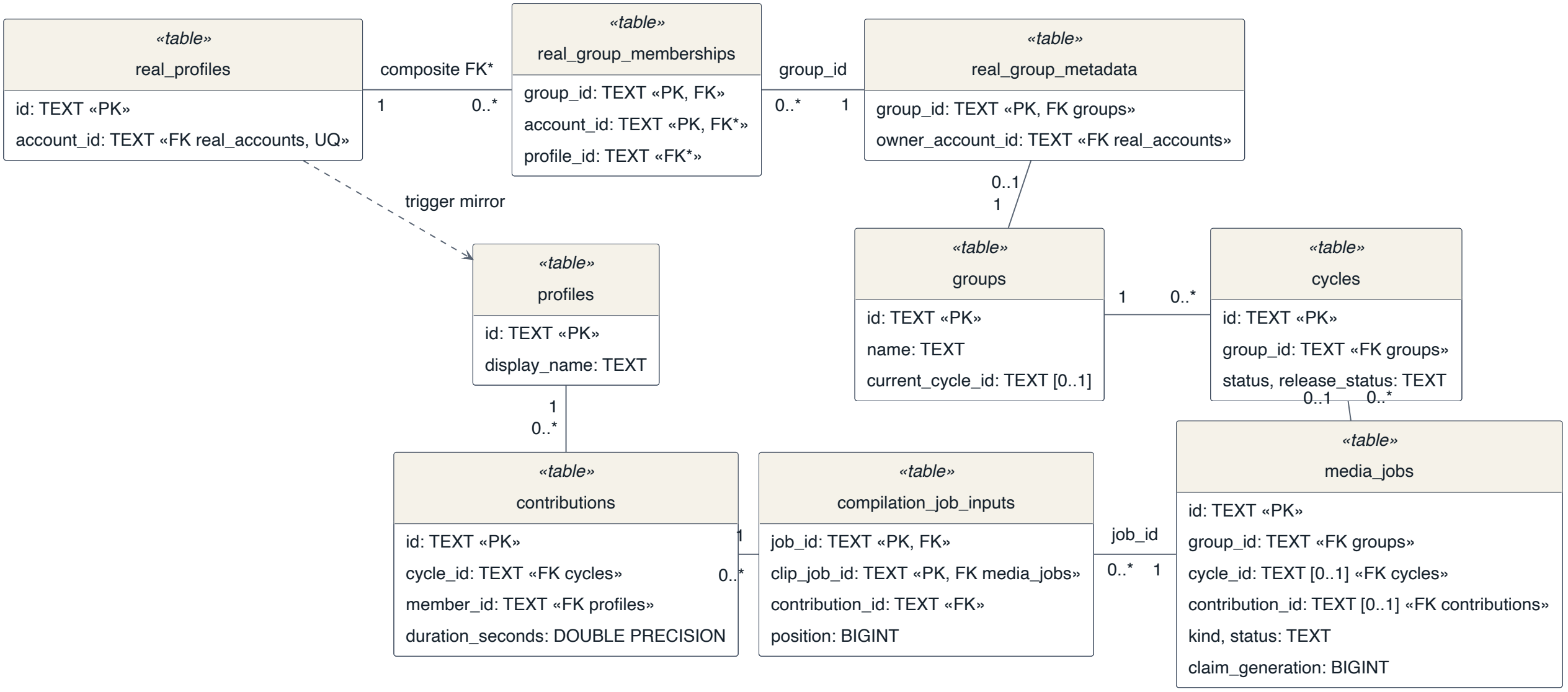
Design sequence · UC07 Close cycle and open successor



Due collecting-cycle path. Continue with seq-cycle-compile and seq-cycle-release.

Keys preserve membership and the film's exact input set

Data class view · Real membership, capture and compilation



Selected columns. PK = primary key; FK = database foreign key; UQ = unique; [0..1] = nullable.
FK* = (account_id, profile_id) → real_profiles(account_id, id). real_accounts is the account identity table.
current_cycle_id is an application link. UNIQUE(job_id, contribution_id). Archive reads published cycles + film jobs.

Each test layer catches a different integration failure

Type	Tool	Count	CI lane	Blocks?	Catches
Static	Prettier / ESLint / tsc	4 checks	Static check matrix	YES	Types, style, imports
Client	Jest / Testing Library	1,674	Frontend + coverage	YES	Screens, state, retry
Server	node:test · SQLite	985 pass 21 skip	Server + coverage	YES	API, privacy, cycles
Database	node:test · PostgreSQL 17	989 pass	PostgreSQL suite	YES	Dialect and schema
Repository / IaC	node:test + Python	123 + 13	Root / IaC	YES	Policy and scan gate
Deployment	Shell / Python fixtures	11 scripts	Cloud-free fixtures	YES	Install and rollback
Browser / a11y	Playwright + Axe	41 / 6 Axe	Web / browser	main	PWA, layout, keyboard
Real-account	Playwright · Firefox	1 flow	Local slow check	NO	Owner, member, outsider
Performance	Node HTTP + FFmpeg	1,200 API / 3 films	Manual performance run	NO	Latency, film, privacy

89.64%
Client statements · floor 70%

92.33%
Client lines

85.67%
Server lines · SQLite

“Try again” must return to a usable viewfinder

Behavioural regression view · CU-A photo retry · #581

GIVEN

Photo capture fails

WHEN

User taps “Try again”

THEN

Viewfinder returns;
a fresh capture succeeds

Before fix		At f6cd896
FAIL	→	PASS

Fault. Retry called capture from a state where its access guard immediately returned.

Repair. Return to the viewfinder so the user can make a fresh capture.

Check. Audit replayed the old source: the new test failed before the fix and passed afterward.

The pipeline connects protected source to a verified deployment

Source → Quality → hosted dev

SWE5006 · TEAM 14 · DEVSECOPS

Every change passes through explicit decisions

Rewind CI/CD · source, security, artifacts and hosted dev

DECISION 1 — QUALITY GATE

Pull request

Focused change → protected dev

DECISION 2 — QUALITY GATE

Parallel verification

Tests · coverage · dependency audit

DECISION 3 — TRIAGE

Security findings

CodeQL · secrets · images · IaC

DECISION 4 — UPLOAD GATE

Verified release bundle

Trivy · SBOM · manifest · provenance

DECISION 5 — HOSTED DEV

Deploy and inspect

OIDC → host health → ZAP baseline

DECISION 6 — REVIEW GATE

Promotion

Protected main → approving review

dev · Quality · signed artifacts · hosted dev /api/health · promotion PR #586

Rewind · CI/CD · hosted dev

A private-group journey runs from prompt to film

01–02

Create and invite

Prompt · private membership

03–06

Contribute

Photo · video · chat · reminder

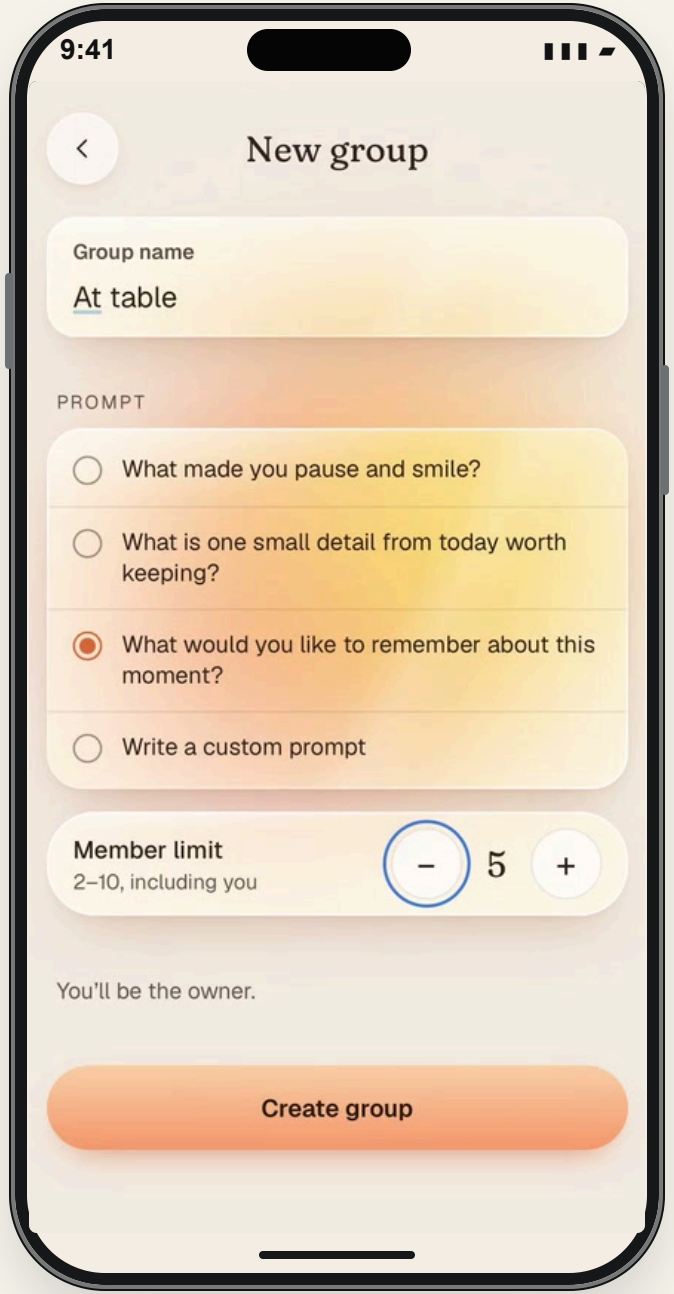
07–10

Reveal and retain

Ready · watch · save · access check

28-day product cycle · private groups · one feature per slide

A shared prompt gives the group a common starting point



01 / 10 · Create a group

DO.

Name the group, set the prompt and choose its size.

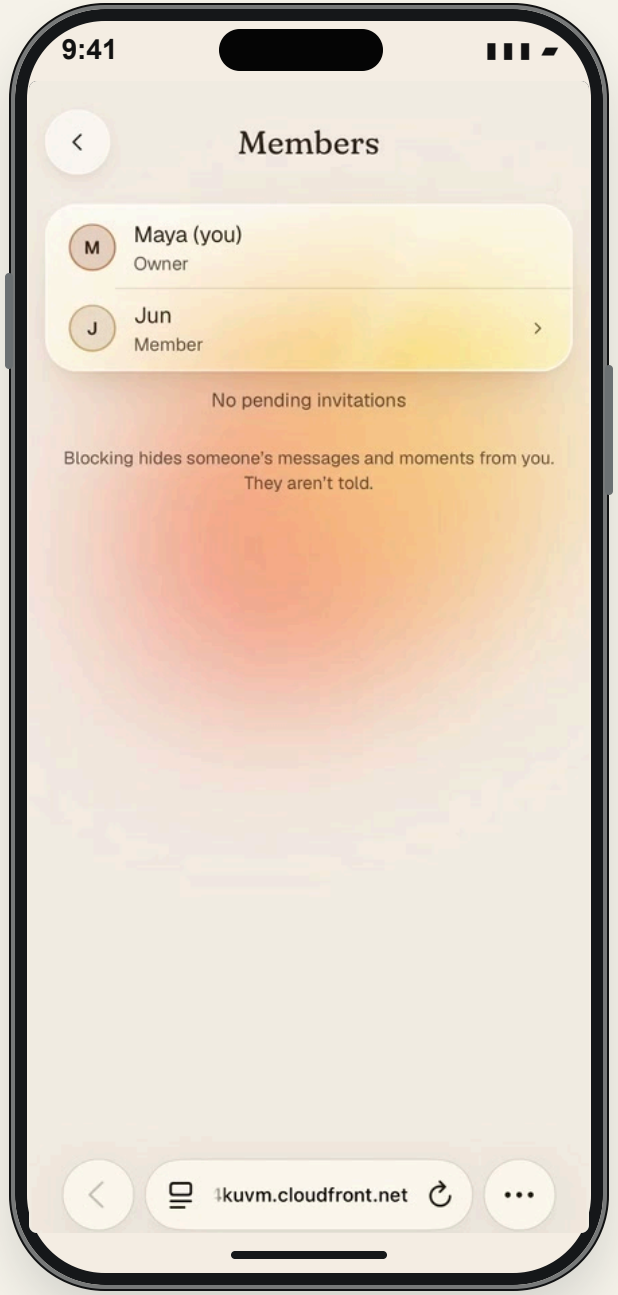
OBSERVE.

Home shows the new group; settings retain prompt and time zone.

RULE.

The owner manages group settings.

One accepted invitation creates one private membership



`02 / 10 · Invite and join`

DO.

Create and share the code; the friend accepts.

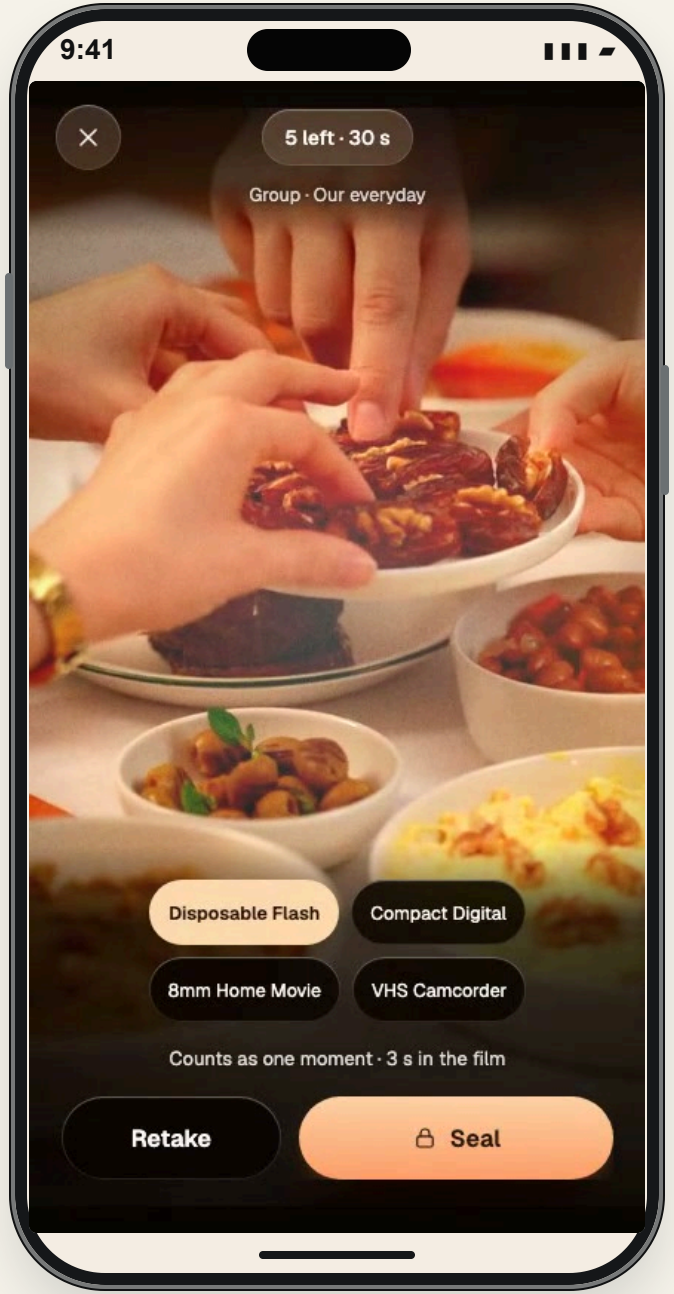
OBSERVE.

Both accounts see the same group and member list.

RULE.

Code works once and expires in 24 hours.

A submitted photo becomes a sealed moment



`03 / 10 · Seal a photo`

DO.

Take a photo, choose a look, review and submit.

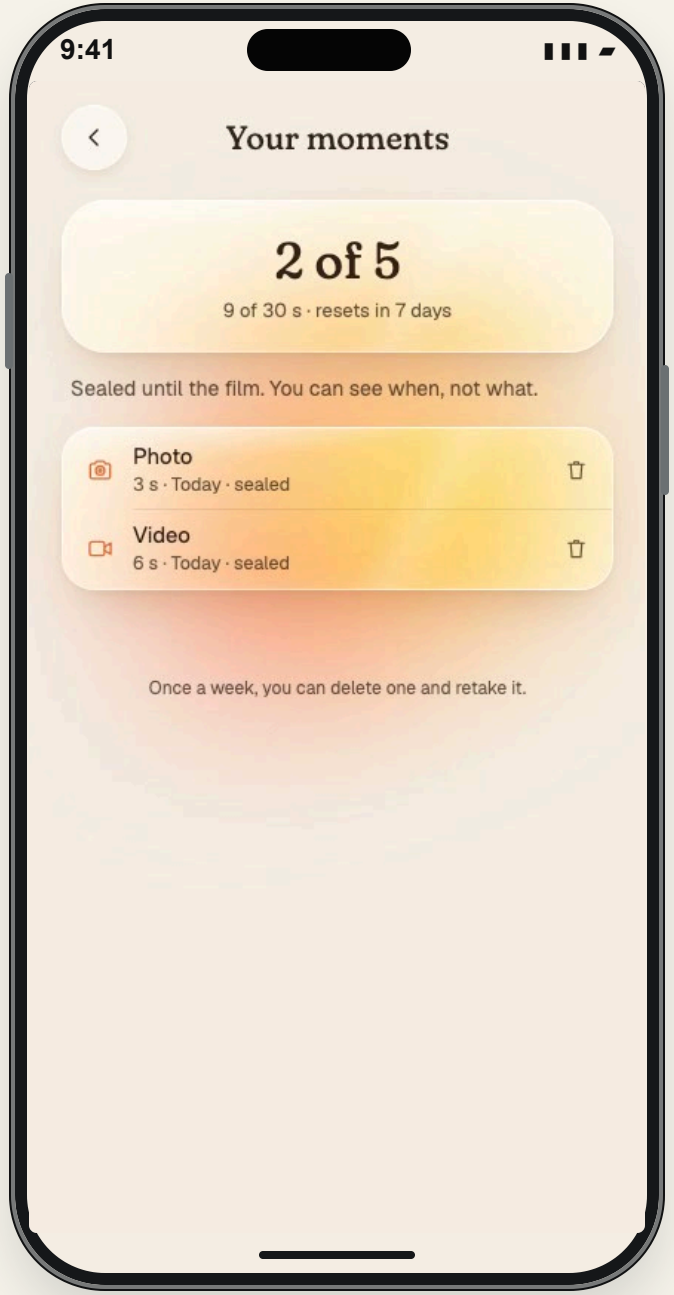
OBSERVE.

Your moments records a sealed entry.

RULE.

One photo uses one moment and three film seconds.

Video and photos share one enforced weekly allowance



04 / 10 · Video and allowance

DO.

Record a video, review and submit.

OBSERVE.

The ledger updates moments and total seconds.

RULE.

5 moments / 30 seconds weekly; each video ≤15 seconds.

Stored messages and live updates keep the group connected



05 / 10 · Group chat

DO.

Send a message, reply and react from the other account.

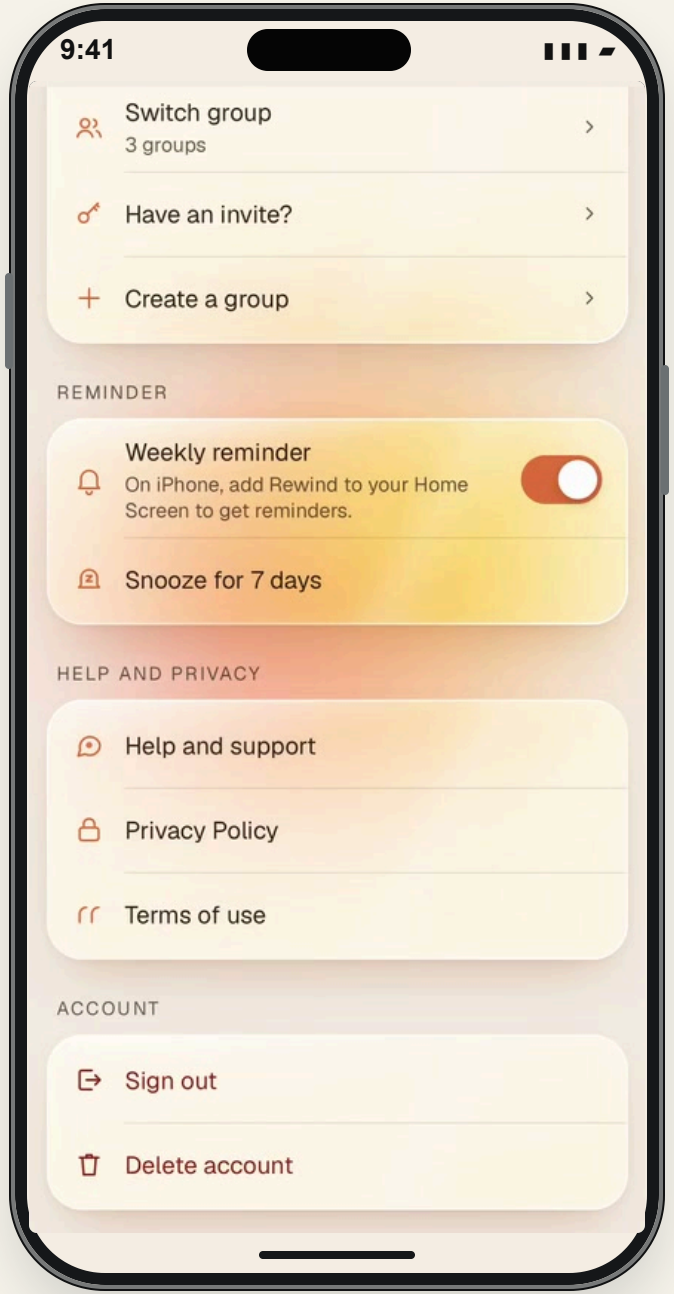
OBSERVE.

The second view updates without a refresh.

DESIGN.

Stored messages plus SSE delivery and reconnect.

Reminder permission leads to an explicit weekly choice



`06 / 10 · Weekly reminder`

DO.

Open settings and choose the weekly reminder.

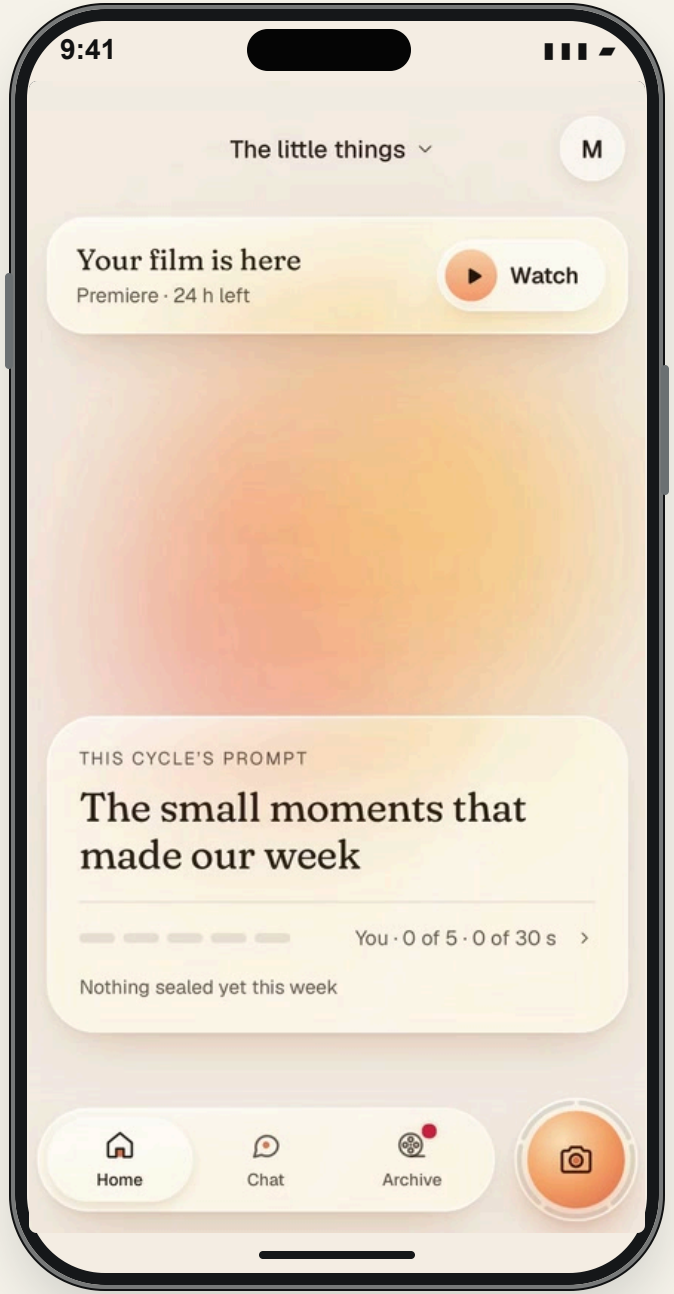
OBSERVE.

The switch and seven-day snooze keep the choice explicit.

RULE.

Sunday 19:00 in the group time zone; iPhone web push uses the Home Screen app.

The completed capsule is ready for its shared premiere



`07 / 10 · Film ready`

OBSERVE.

A completed capsule exposes its premiere card.

DESIGN.

The scheduler closes the cycle, queues work and opens its successor.

RULE.

The product cycle is 28 days.

The reveal turns sealed contributions into one shared film



08 / 10 · Watch the film

DO.

Open the premiere and watch; skip between moments.

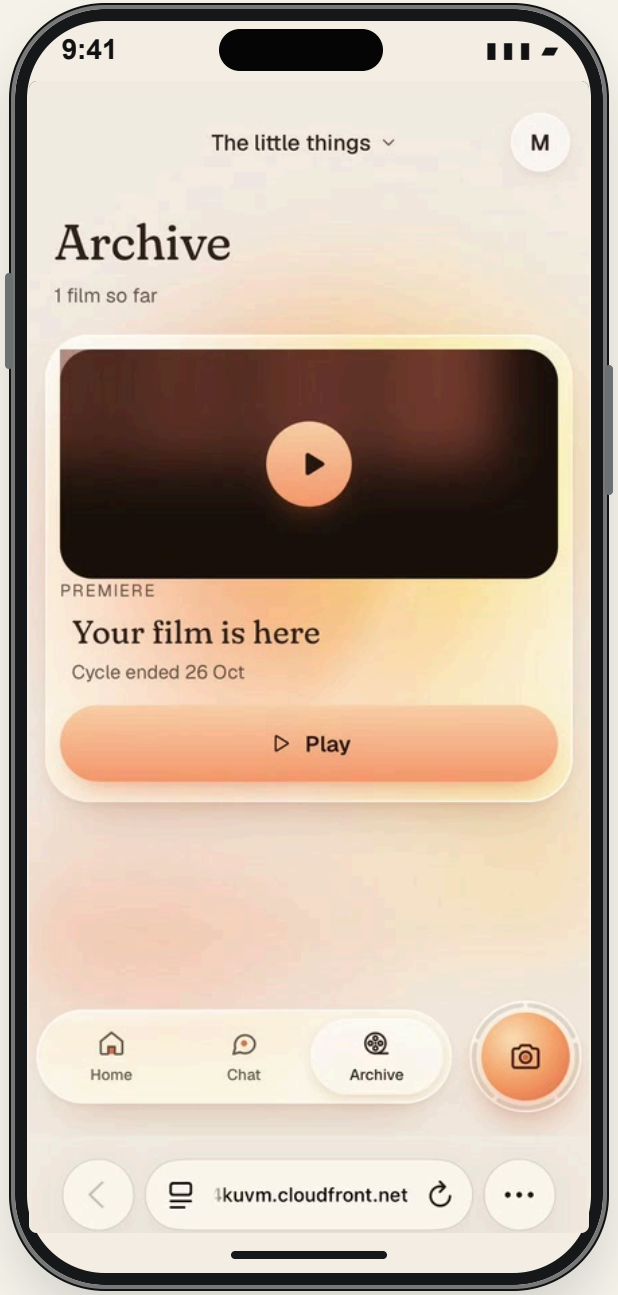
OBSERVE.

One film combines the group’s accepted inputs.

OUTPUT.

720 × 1280 · 30 fps · normalised audio.

Archive keeps the film available after its premiere



09 / 10 · Archive and save

DO.

Open Archive, select the film and save it.

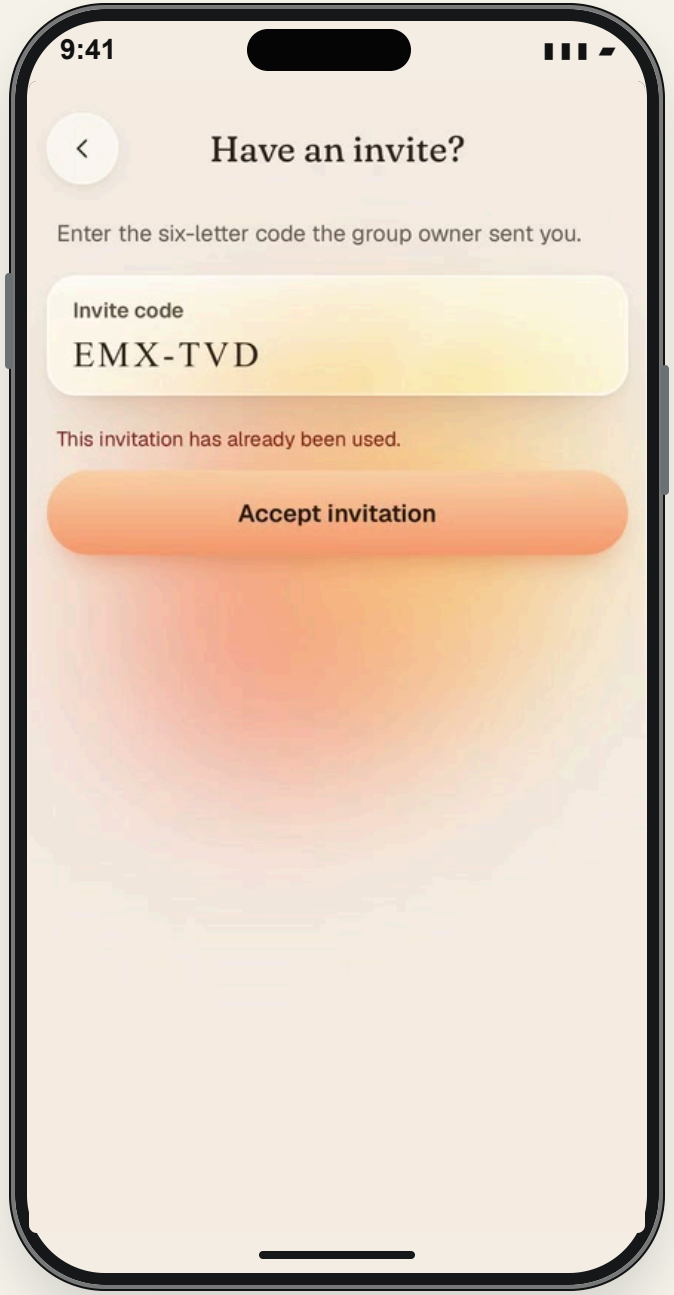
OBSERVE.

The MP4 download opens as a playable file.

CONTROL.

Members can save the film and their own moments.

A spent invitation gives an outsider no group access



`10 / 10 · Outsider denied`

DO.

Reuse the spent invitation from an outsider account.

OBSERVE.

The invitation is rejected and the private group stays absent.

BOUNDARY.

Private group and media reads require their access checks.

Load results identify the next scaling change

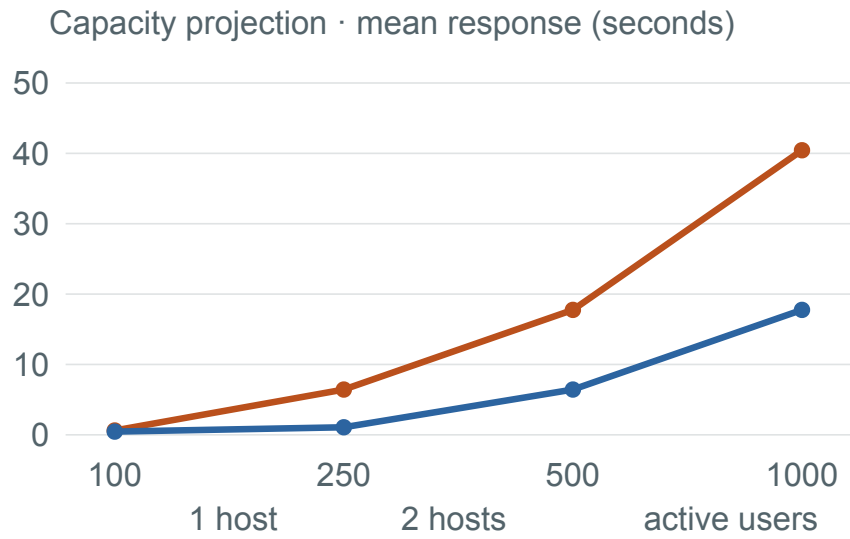
Measured hosted mixed workload · 27 Oct

VUs	p95 ms	Errors	req/s
1	1,058	0.00%	0.21
10	663	0.00%	1.83
25	675	0.00%	4.59
50	842	0.00%	9.06
100	1,336	0.00%	17.84

Method: Node HTTP · 1→100 VUs · 180 s steady per stage · 3–7 s think time.
Gate: p95 <2 s and errors <1%.

> **All five stages pass**
> 6,038 steady requests · 0 errors
> 100 VUs: 1.34 s p95 · 17.84 req/s

Projected capacity · fitted to the quiet-host run



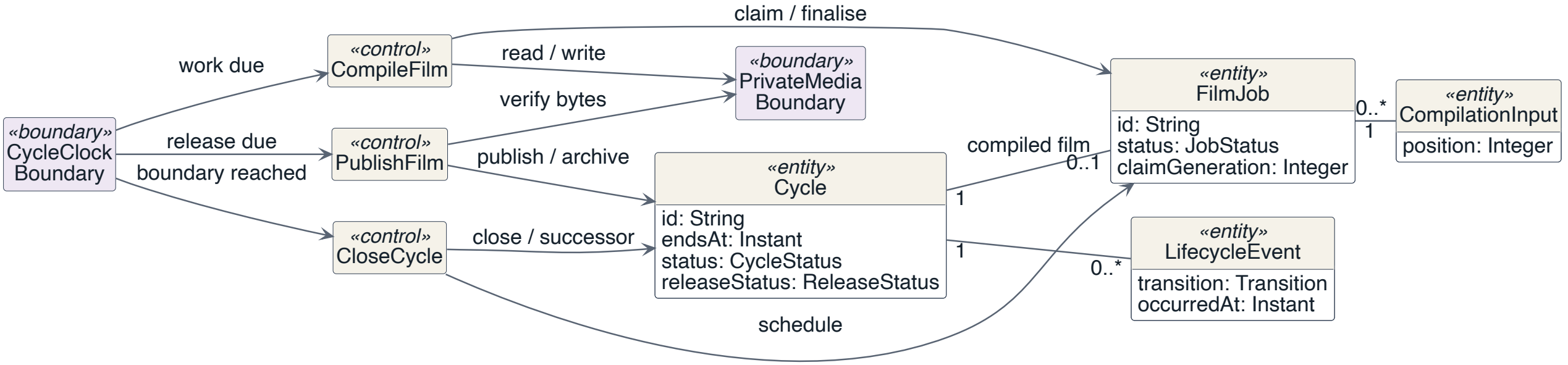
Measured. 100 users leave 0.66 s below the 2 s p95 target.

Scale. Separate media work; remove synchronous database serialization; coordinate shared state.

Maintain. The same pipeline checks the next change.

Analysis separates interaction, coordination and state

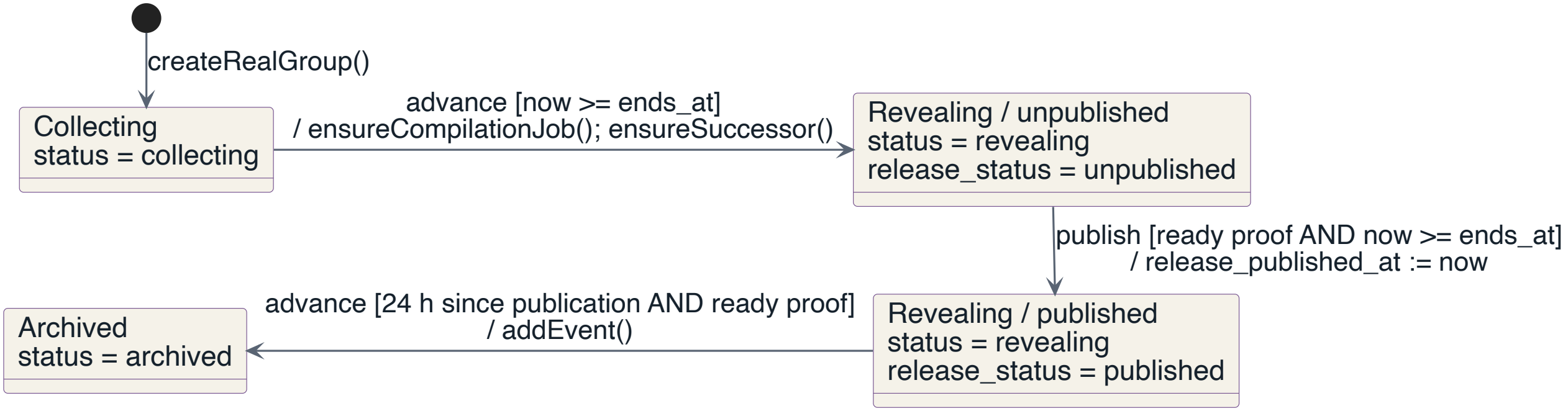
Analysis classes · UC07-UC08 Close cycle and release film



Conceptual BCE responsibilities; mapped to implemented elements in a2d-cycle.

Durable state survives interrupted compilation

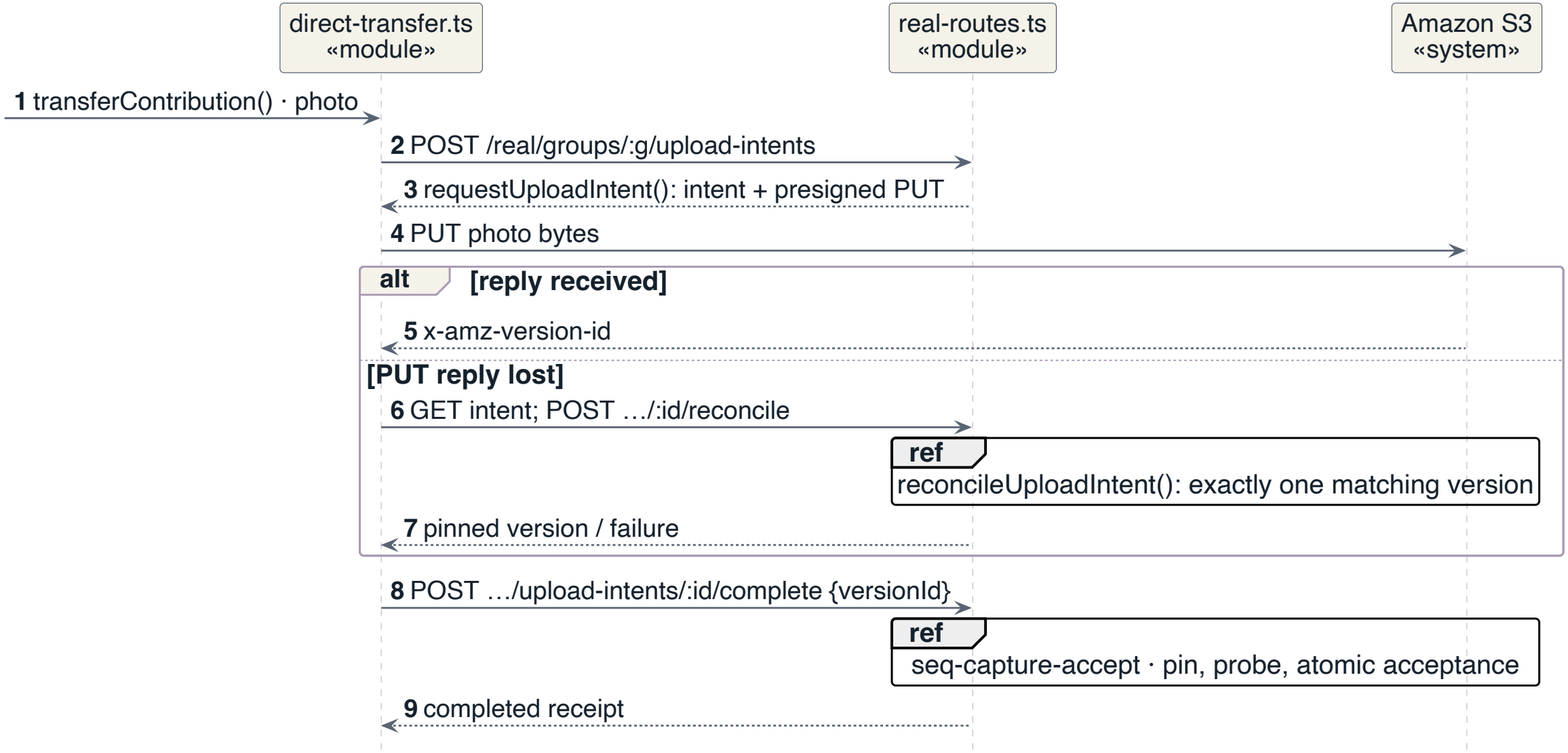
Statechart · UC07-UC08 Close cycle and release film



advance = advanceCycleLifecycleWithStore(); publish = publishCycleReleaseWithStore(). Ready proof = verified film bytes + unchanged ready job. Failed guards retain the state. Intake closes at ends_at via cyclePhase(). The successor collects during release.

Capture acceptance checks identity, allowance and media

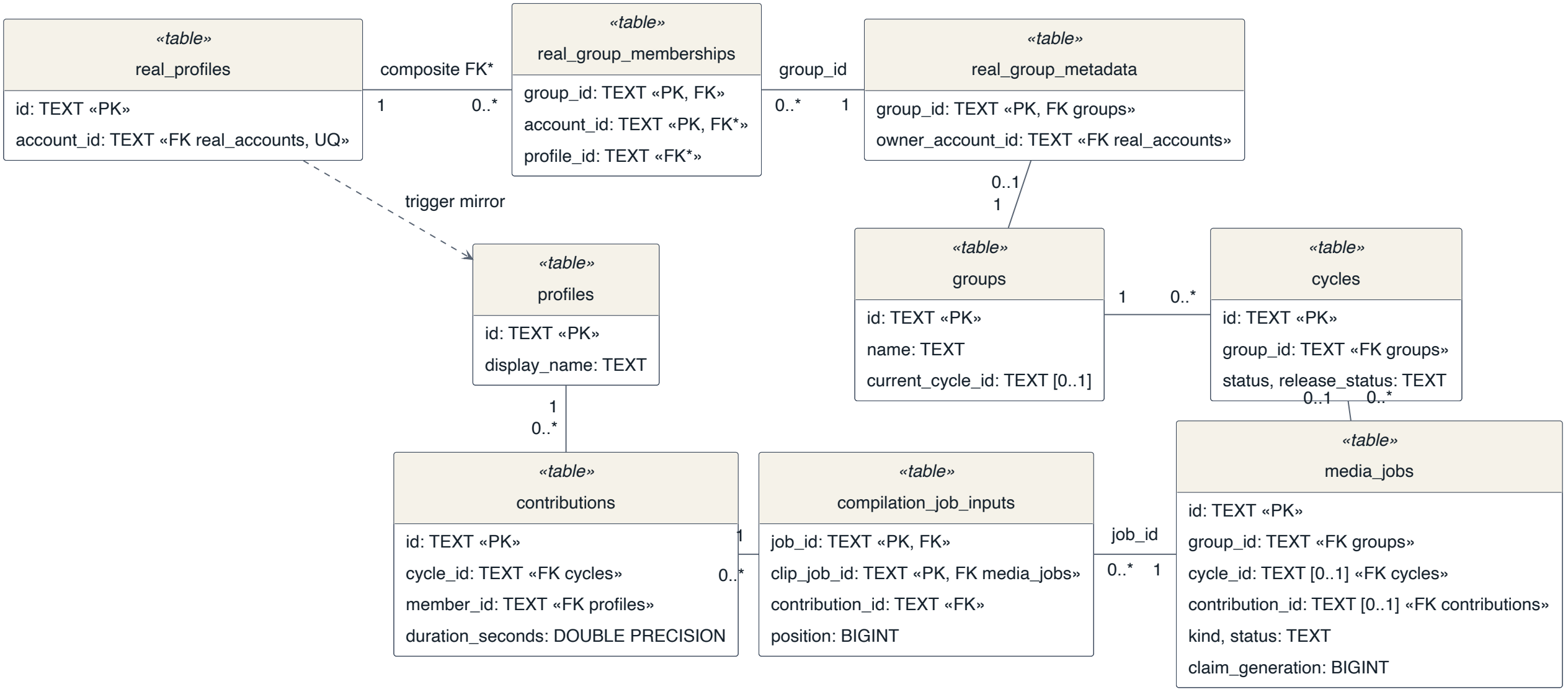
Design sequence · UC05-UC06 Contribute a photo within allowance



CameraCaptureScreen → GroupCaptureView callback → submitPhotoContribution() → runtime uploadClip() → transferContribution(). Processing: seq-capture-process.

Relational keys make the private film traceable

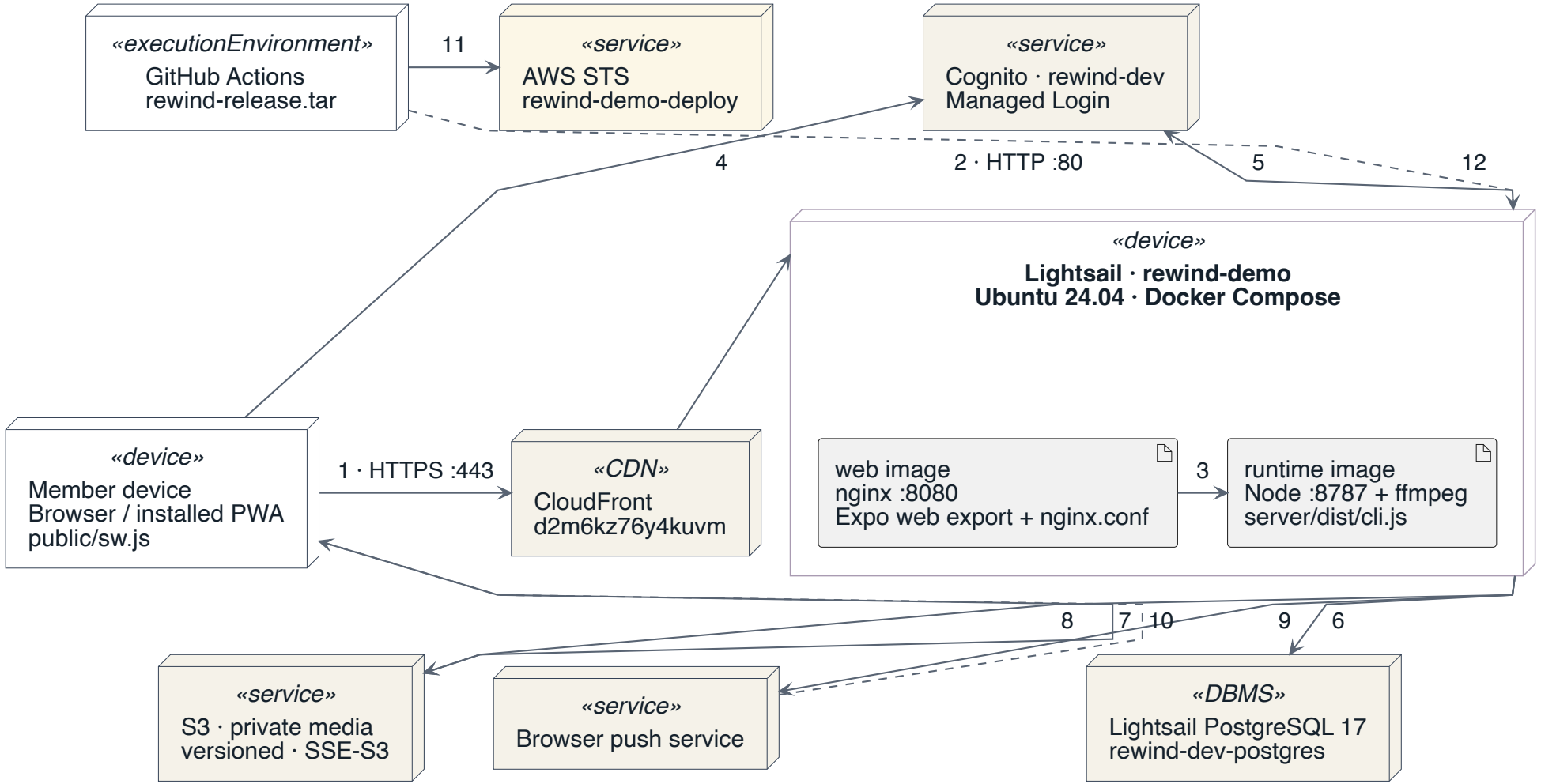
Data class view · Real membership, capture and compilation



Selected columns. PK = primary key; FK = database foreign key; UQ = unique; [0..1] = nullable.
FK* = (account_id, profile_id) → real_profiles(account_id, id). real_accounts is the account identity table.
current_cycle_id is an application link. UNIQUE(job_id, contribution_id). Archive reads published cycles + film jobs.

The deployment view names each runtime boundary

Deployment view · Hosted dev



1 Browser → CloudFront: HTTPS :443. 2 CloudFront → nginx: HTTP :80 → :8080 + origin auth header.
3 nginx → Node: HTTP :8787 /api + SSE. 4 Browser → Cognito: HTTPS :443, authorization code + PKCE.
5 Node → Cognito: HTTPS :443 token exchange. 6 Node → PostgreSQL: TLS :5432.
7 Browser → S3: HTTPS :443 presigned PUT. 8 Node → S3: HTTPS :443 with IAM user credentials.
9 Node → push service: HTTPS :443 / VAPID. 10 Push service → PWA: push event.
11 Actions → STS: OIDC / HTTPS :443, 1 h dev role. 12 Actions → host: SCP / SSH :22, runner /32.
AWS ap-southeast-1. OIDC: dev branch + environment. Node owns the cycle, media and reminder loops.

Coverage and enforcement answer different questions

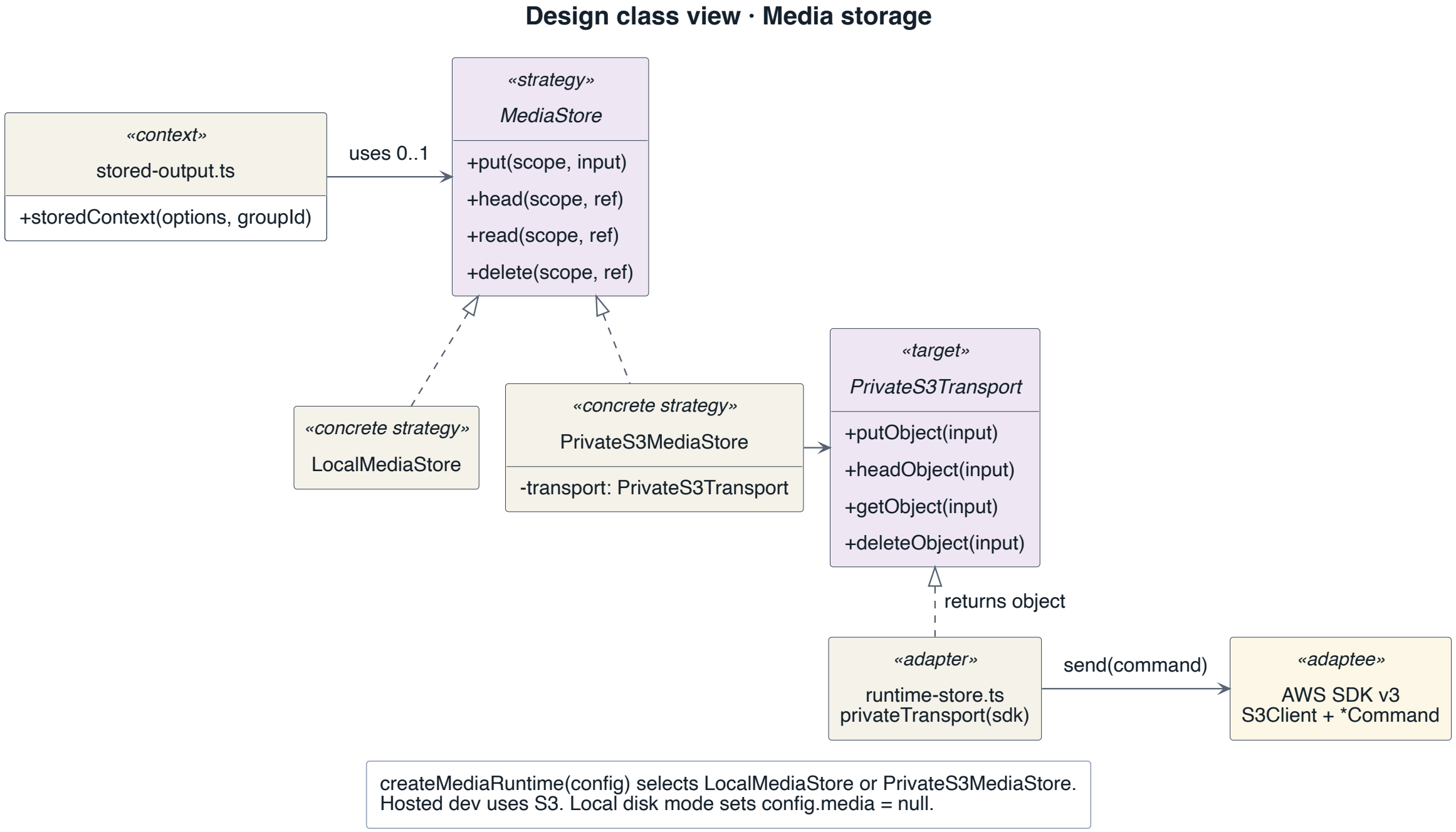
Coverage	Local 26 Oct	CI 20 Oct	Policy
Client statements	89.64%	89.66%	70% statement floor
Client branches	85.68%	85.66%	Reported
Client functions	89.55%	89.61%	Reported
Client lines	92.33%	92.33%	Reported
Server lines	85.67%	85.50%	Reported
Server branches	85.62%	85.66%	Reported
Server functions	85.05%	86.41%	Reported

Cases. 1,674 frontend; 985 server pass + 21 skip; 123 root pass + 1 skip; 41 browser / 6 Axe.

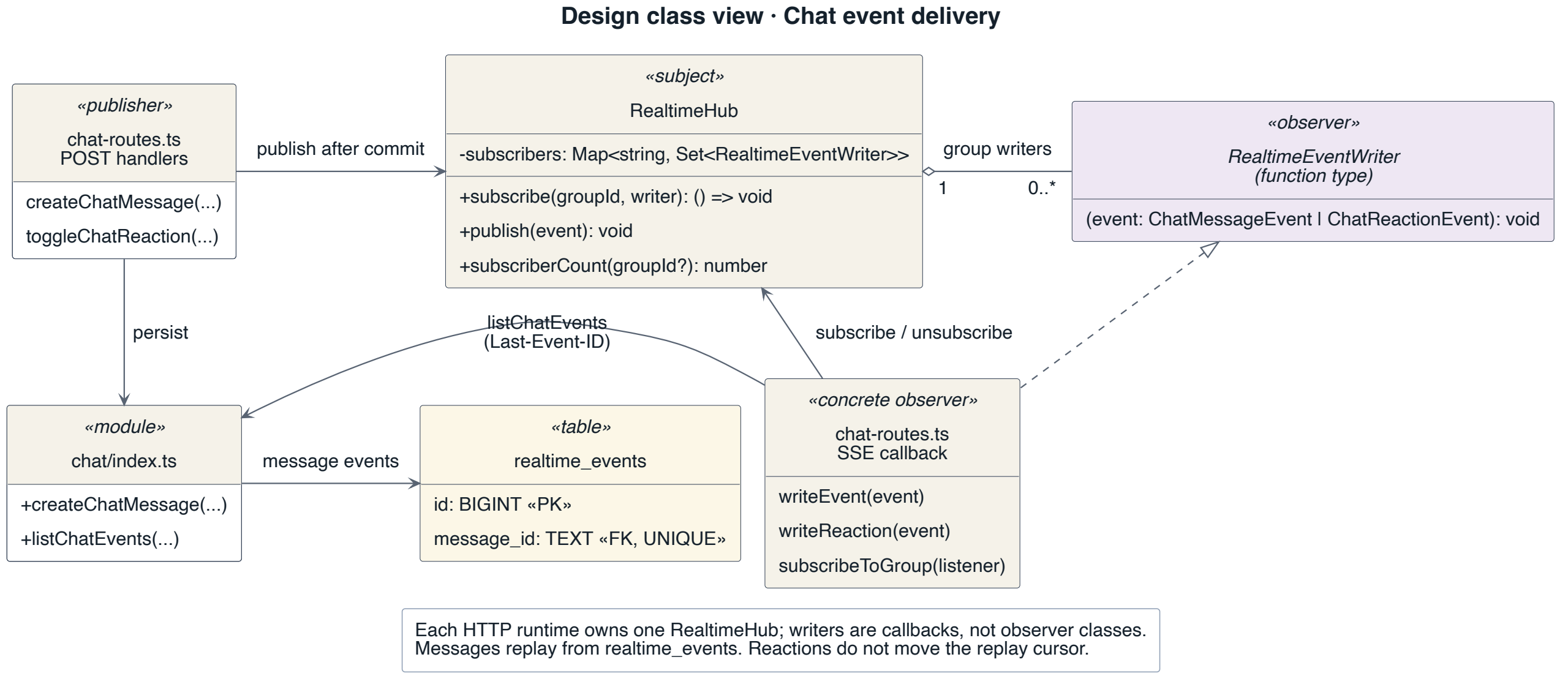
Privacy. Route contracts and outsider tests exercise group, budget, chat and private media denials.

Performance. Local: 1,200 requests; 3 × 25-clip films in 4.9–8.1 seconds.

Storage changes behind Strategy and Adapter roles

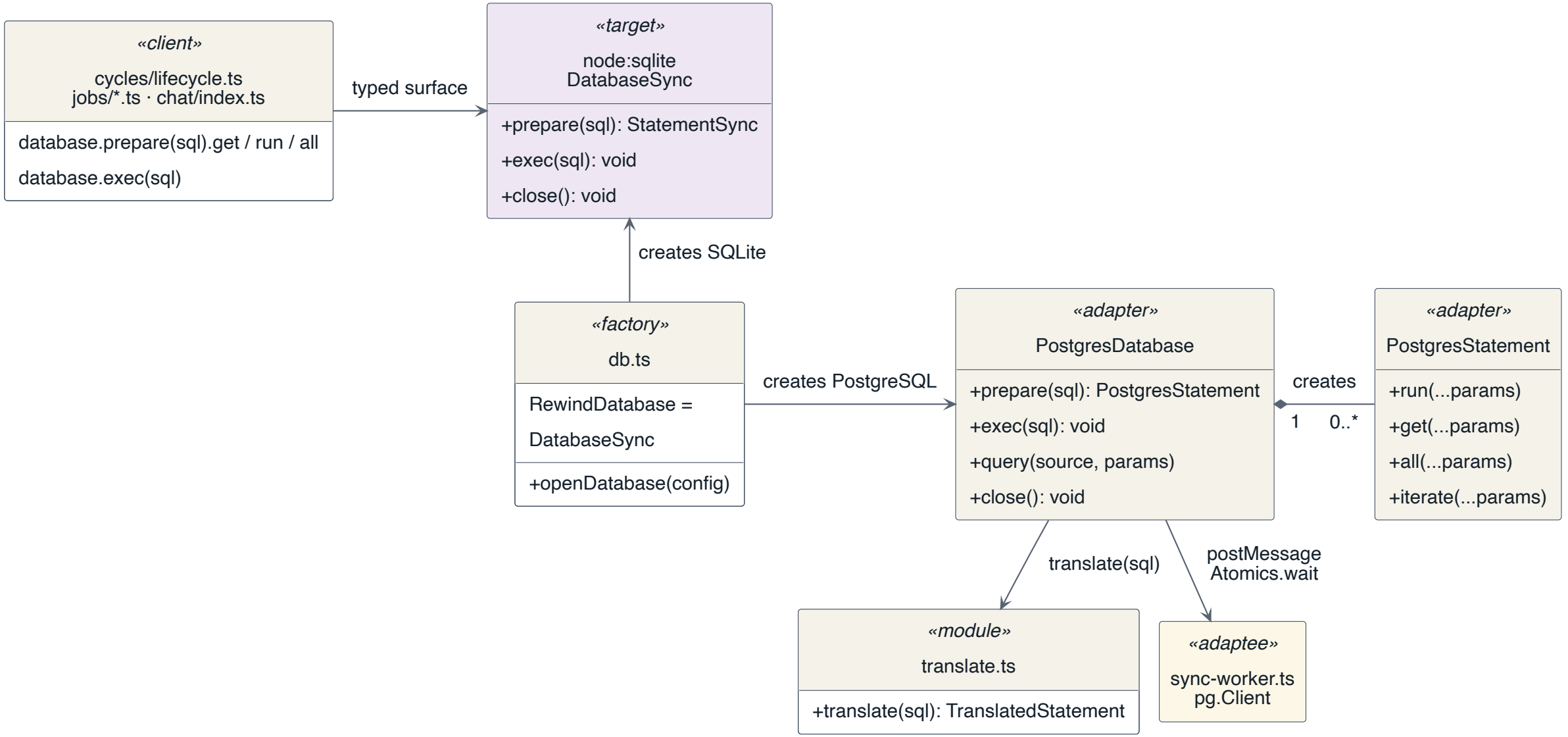


Observer distributes chat updates alongside stored history



The persistence adapter preserves a common data-access contract

Design class view · PostgreSQL persistence adapter

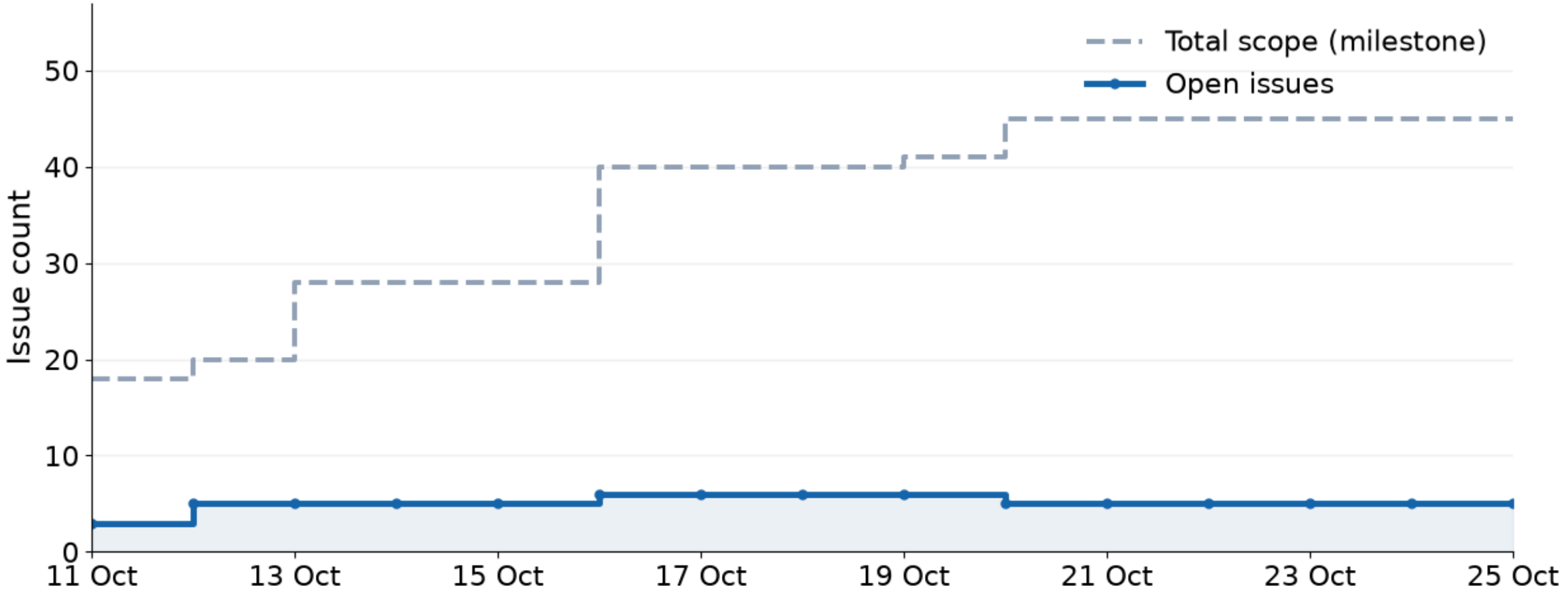


db.ts casts PostgresDatabase to RewindDatabase; the adapter has matching methods. Worker-thread pg.Client performs PostgreSQL I/O; callers keep synchronous operations.

Agile records retain scope changes and equal effort

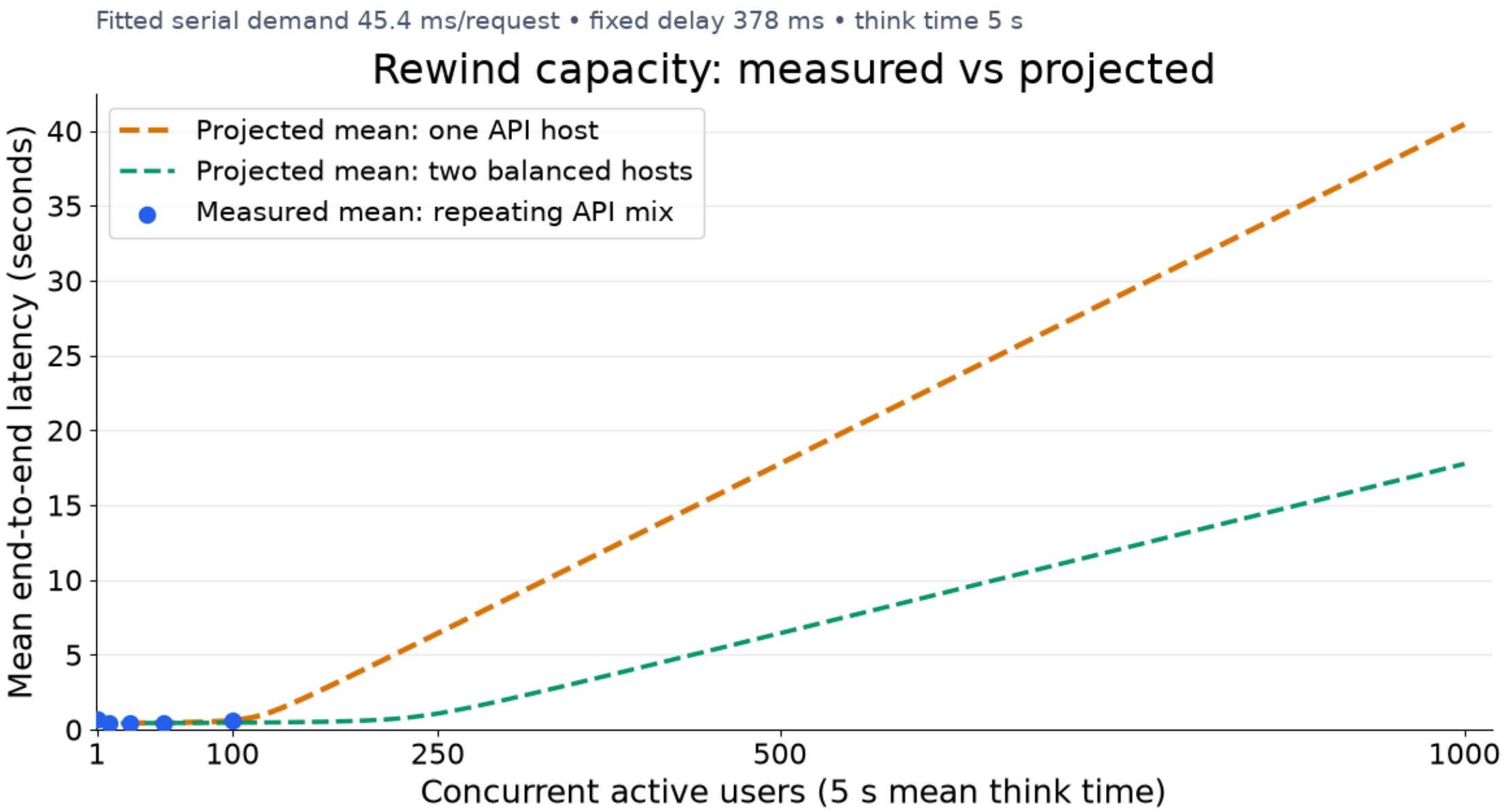
Sprint 3 · issue burndown

11 Oct–24 Oct 2026 · opening 00:00 and daily closes · SGT



Opening scope/open: 18/3 | Closing scope/open: 45/5

The capacity curve follows the closed-user workload



Retained artifacts make the deployment independently inspectable

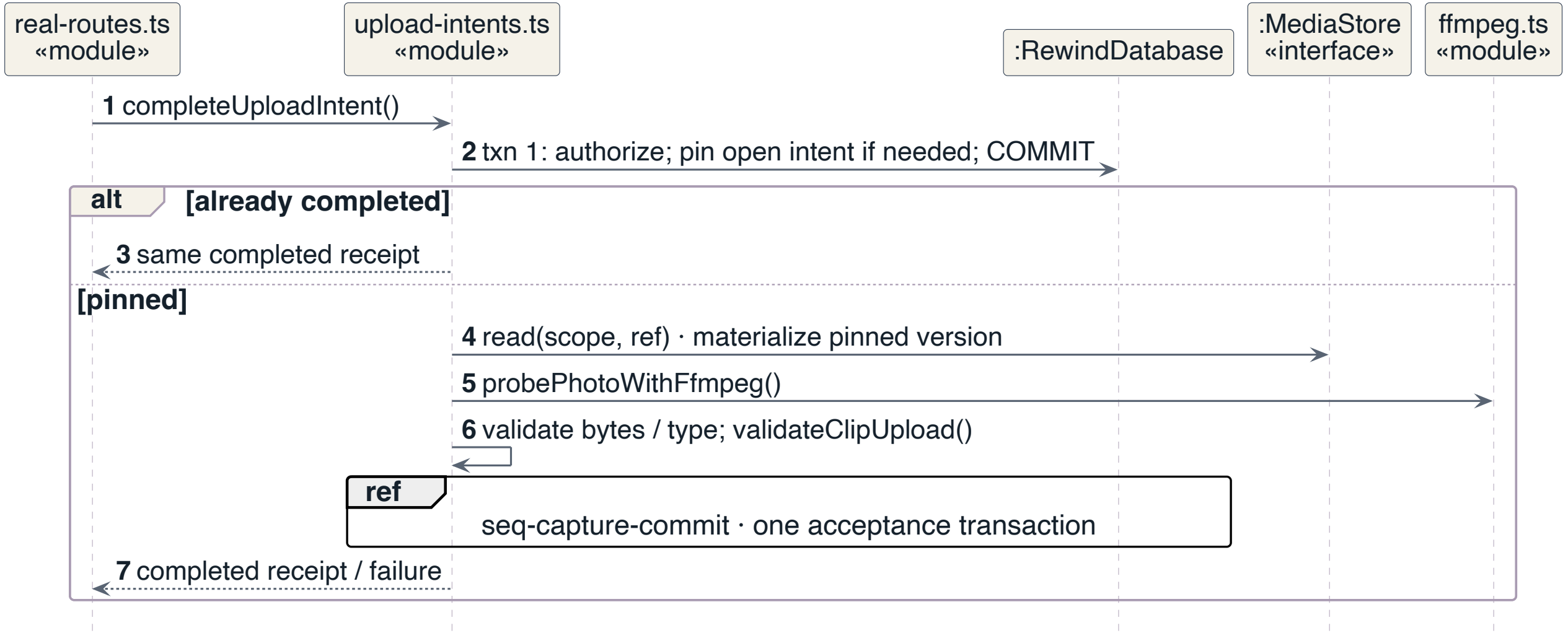
File / output	Question it answers
quality-test-results-root · server/frontend coverage	Which behaviours and coverage were exercised?
npm-audit.json · gitleaks.sarif · iac-full.json	Which dependency, secret and configuration findings were reported?
sbom-runtime.cdx.json · sbom-web.cdx.json	Which components are inside the image pair?
trivy-runtime.json · trivy-web.json	Which findings belong to those exact images?
release-manifest.json · rewind-release.tar	Which commit, bytes and report hashes travelled together?
Signed provenance · gh attestation verify	Which workflow produced the retained subjects?
ZAP report_html.html · report_json.json	Which running-page findings were observed?

```
gh attestation verify --repo Collaboration95/rewind-app release-manifest.json
```

IaC JSON · 15 findings Signed manifest JSON ZAP JSON

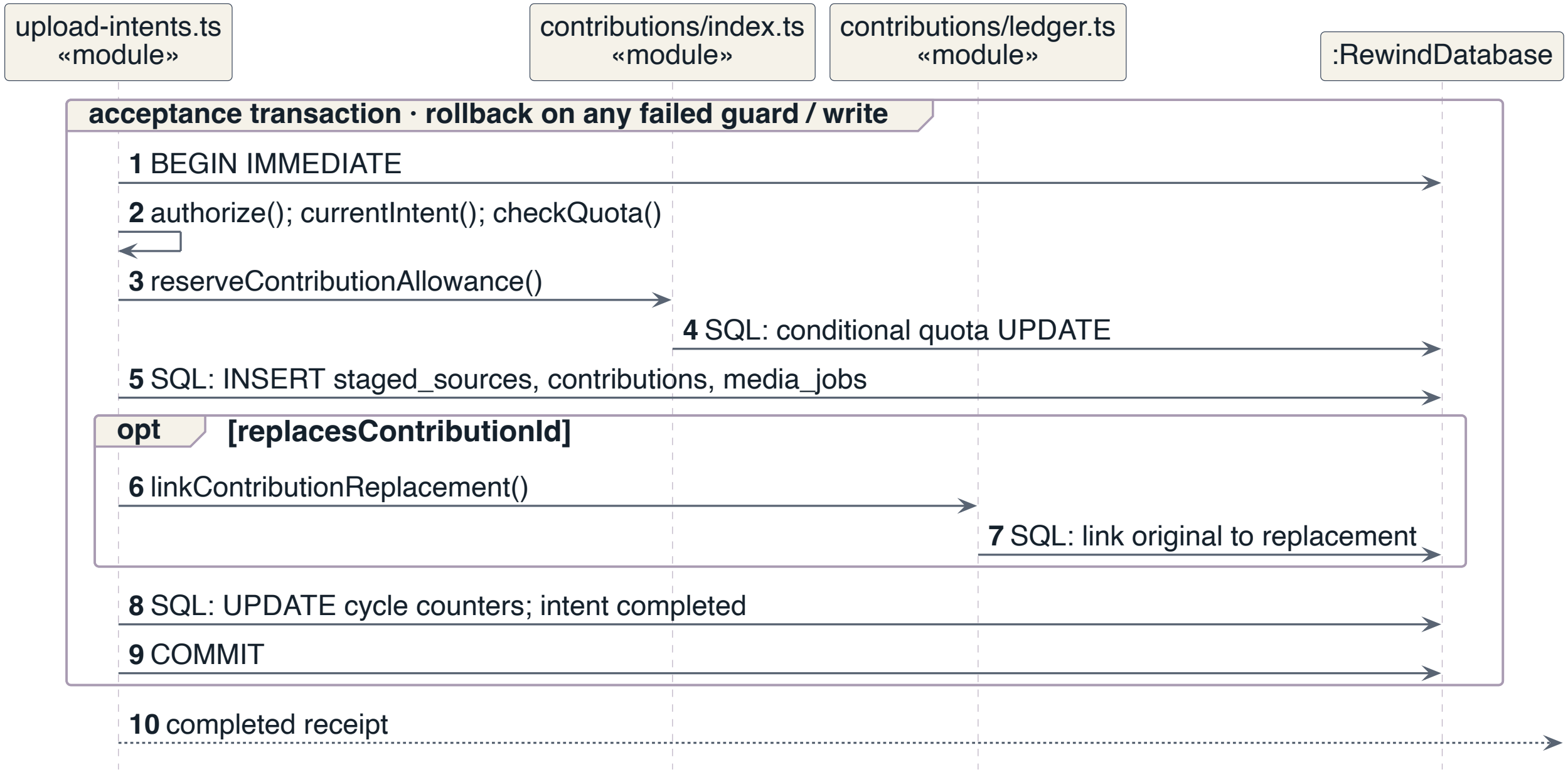
Acceptance pins and probes the exact media version and replays an existing receipt

Design sequence · UC05-UC06 Pin and validate the photo



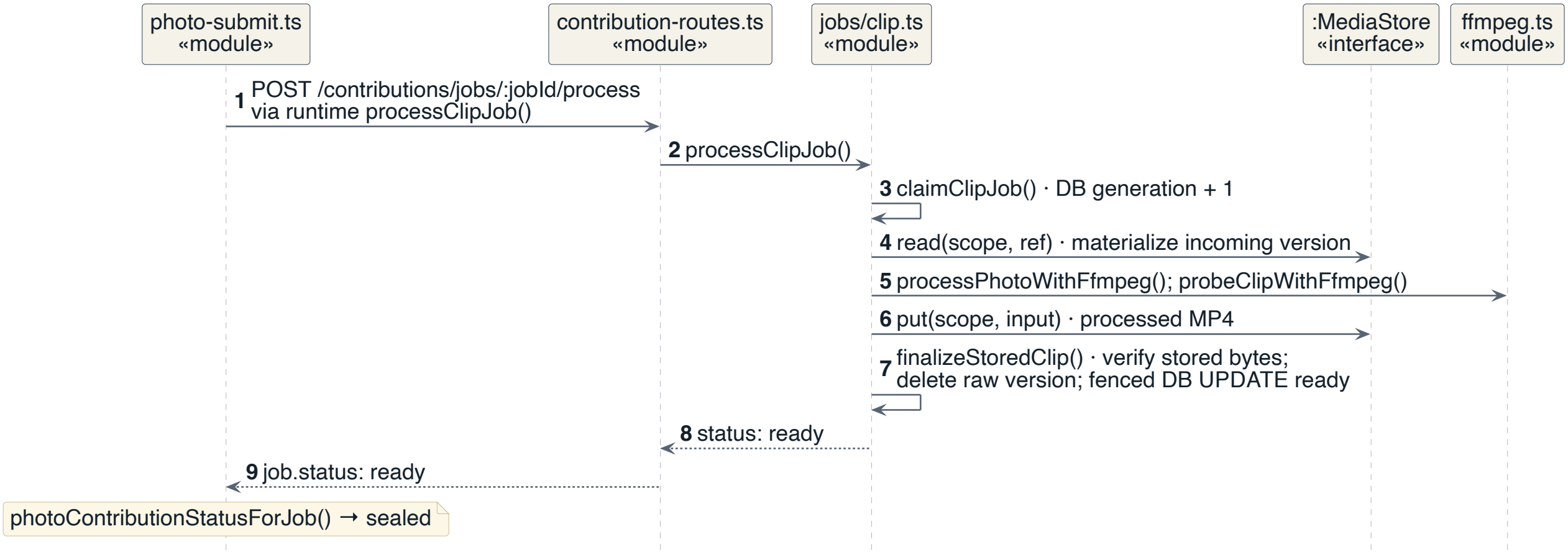
One transaction records quota, contribution, processing job and completion

Design sequence · UC06 Commit the allowance and contribution



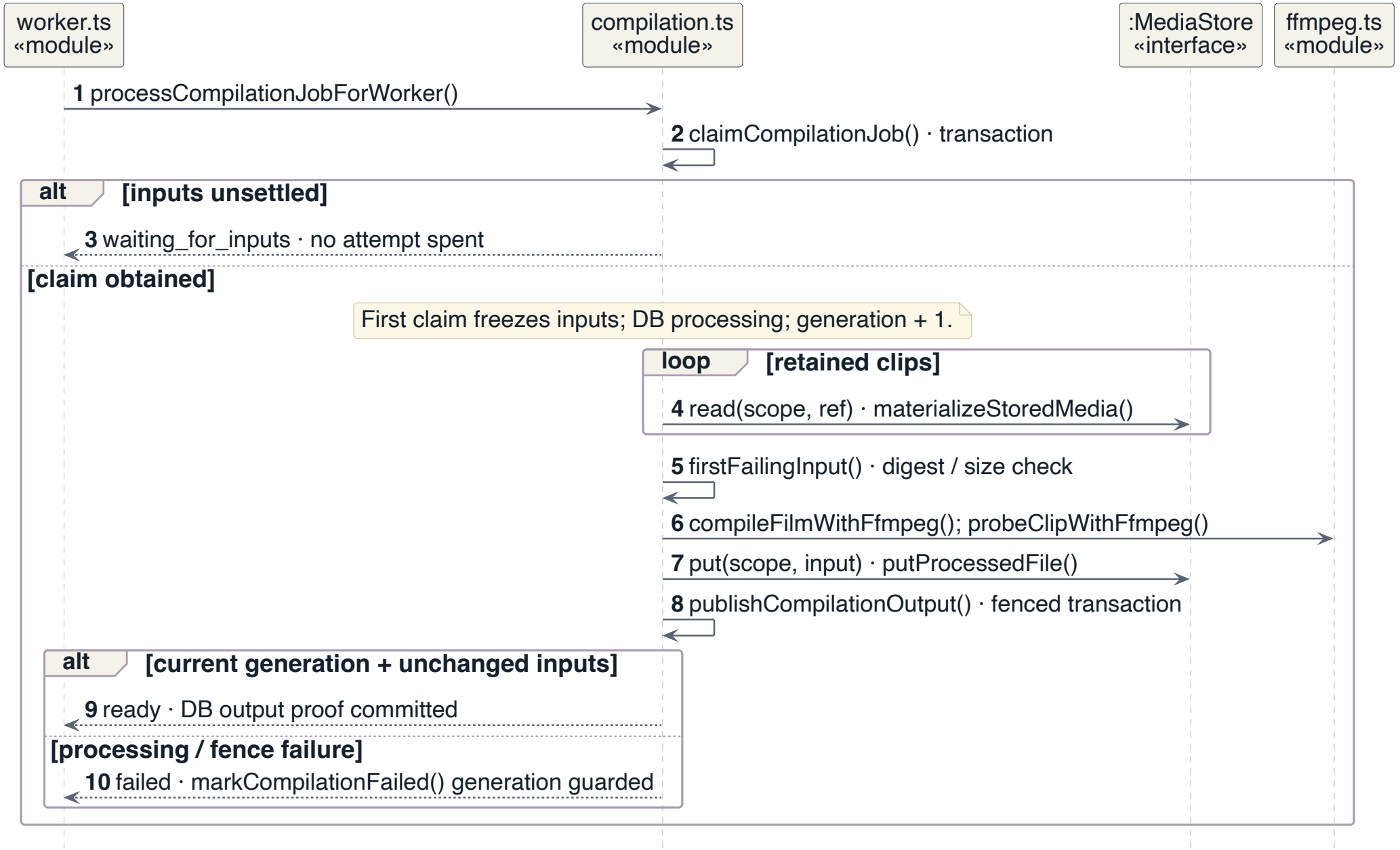
Processing stores an MP4 and proves current ownership before marking ready

Design sequence · UC05 Process the accepted photo



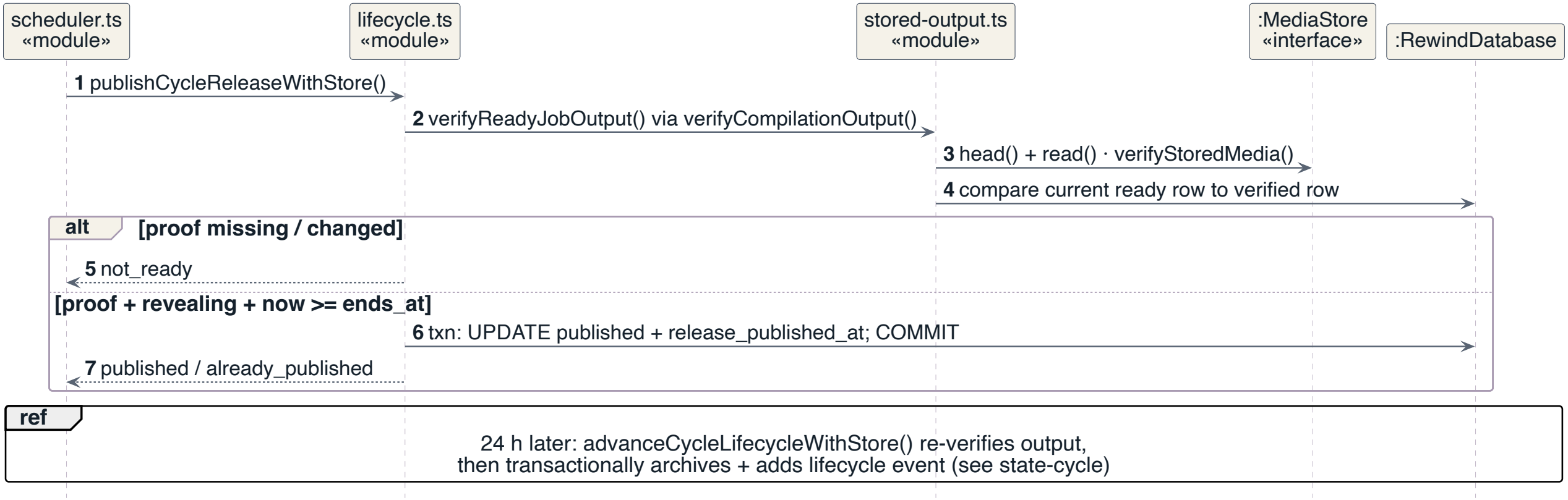
The worker claims settled inputs, materialises media and fences finalisation

Design sequence · UC08 Compile the cycle film



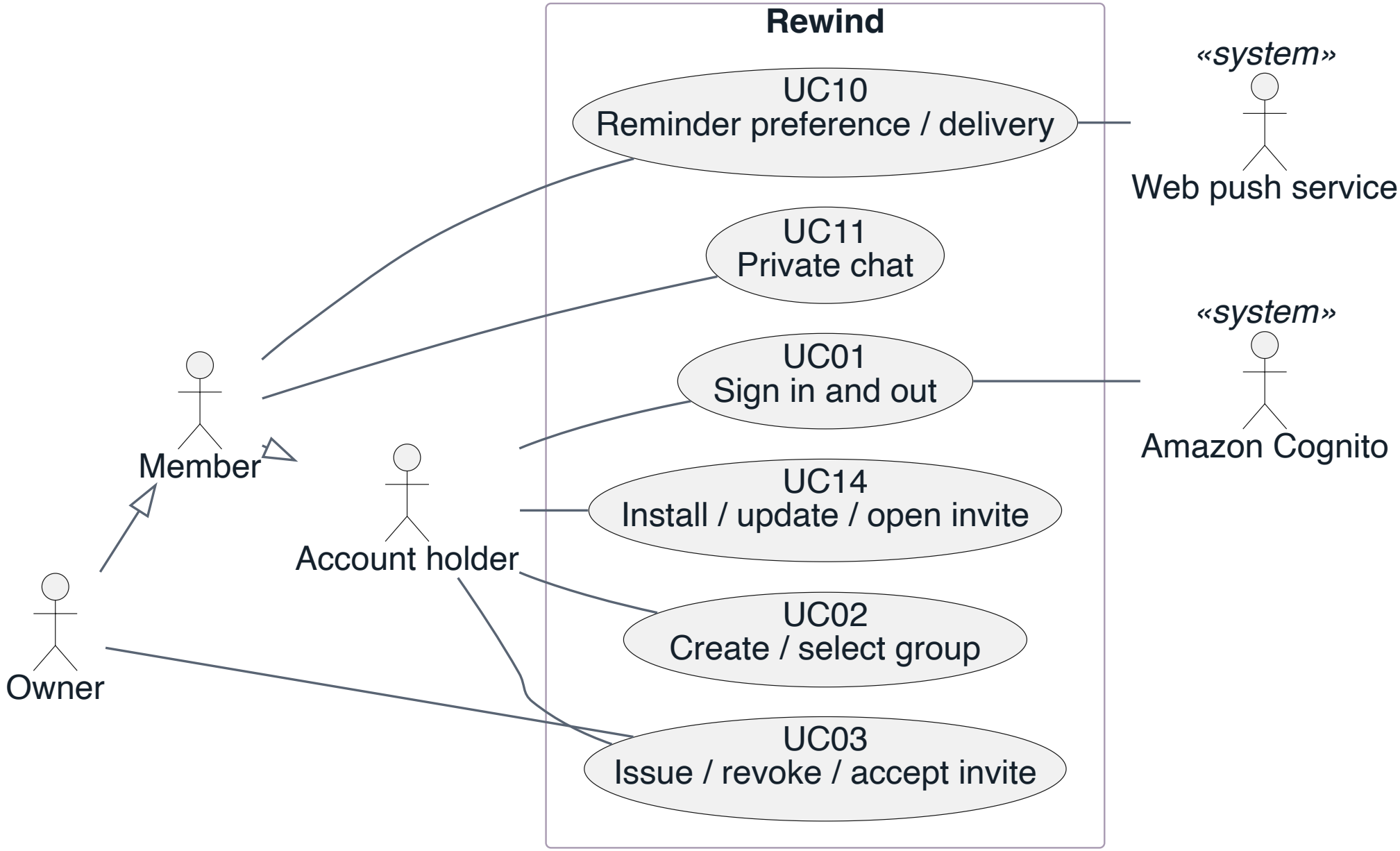
Stored bytes and the current ready row are verified before publication and archiving

Design sequence · UC08 Publish and archive the film

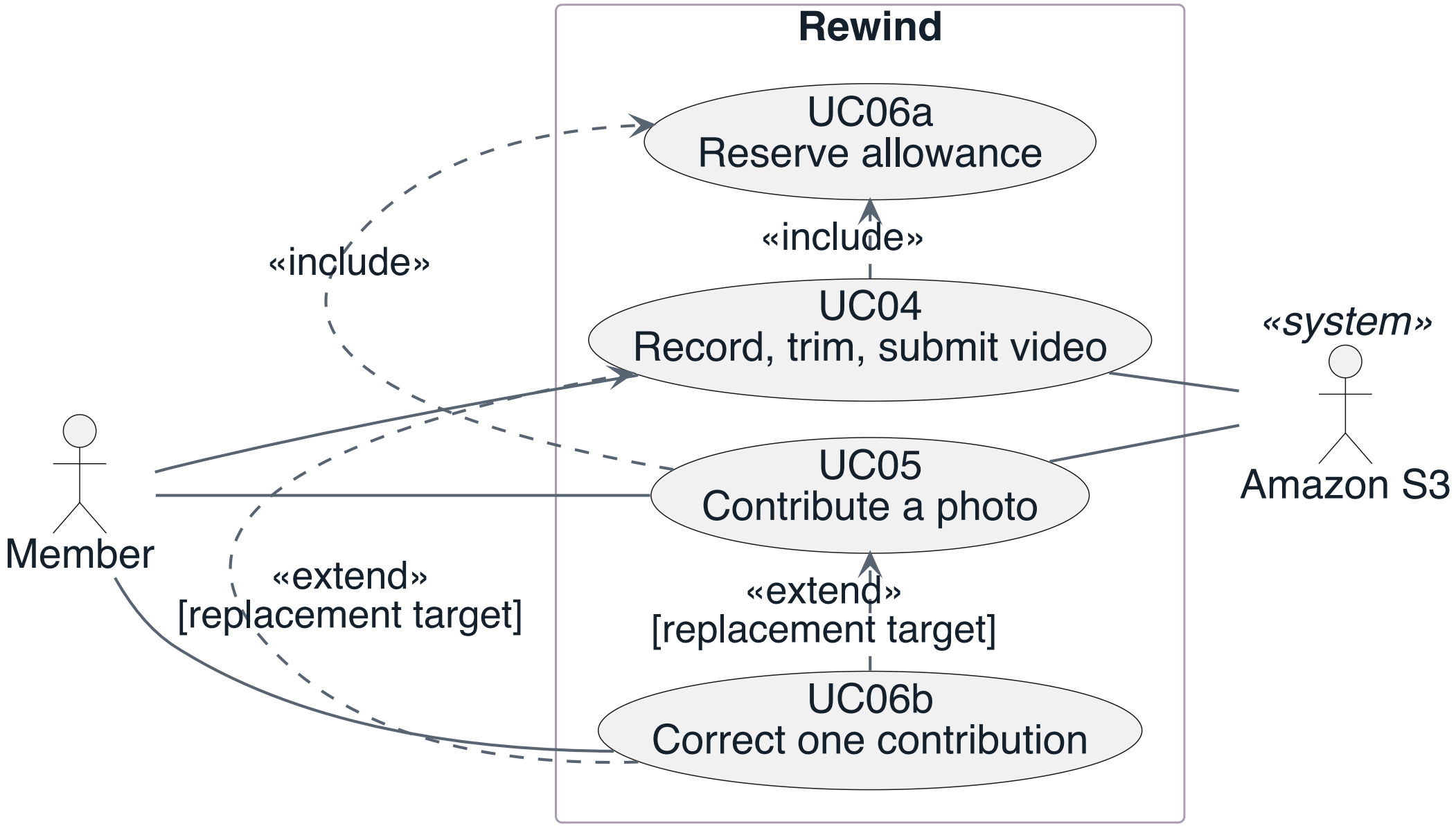


Account and group goals connect to Cognito, chat and web push

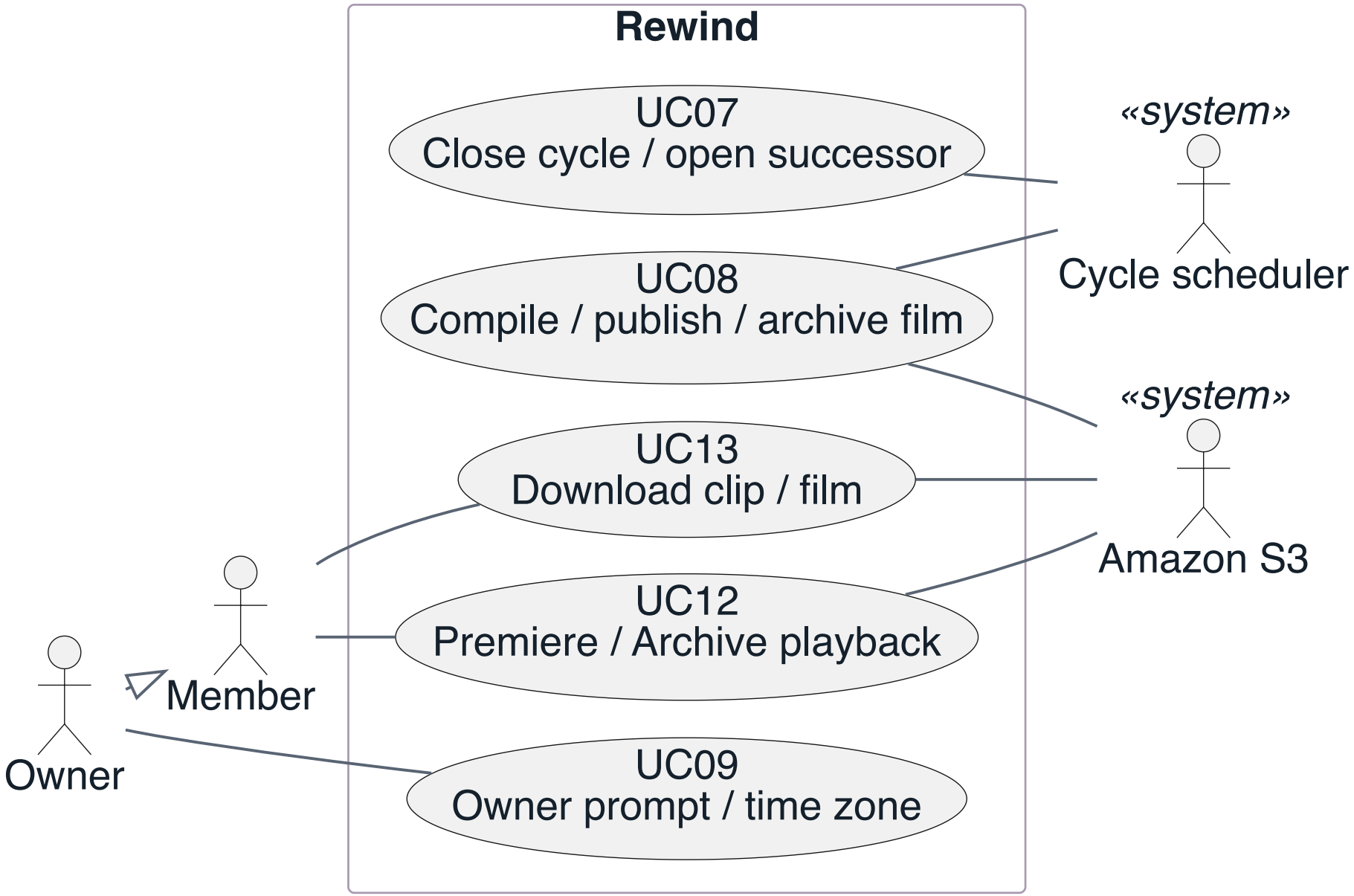
Use case model · UC01-UC03, UC10-UC11, UC14 Participate in a group



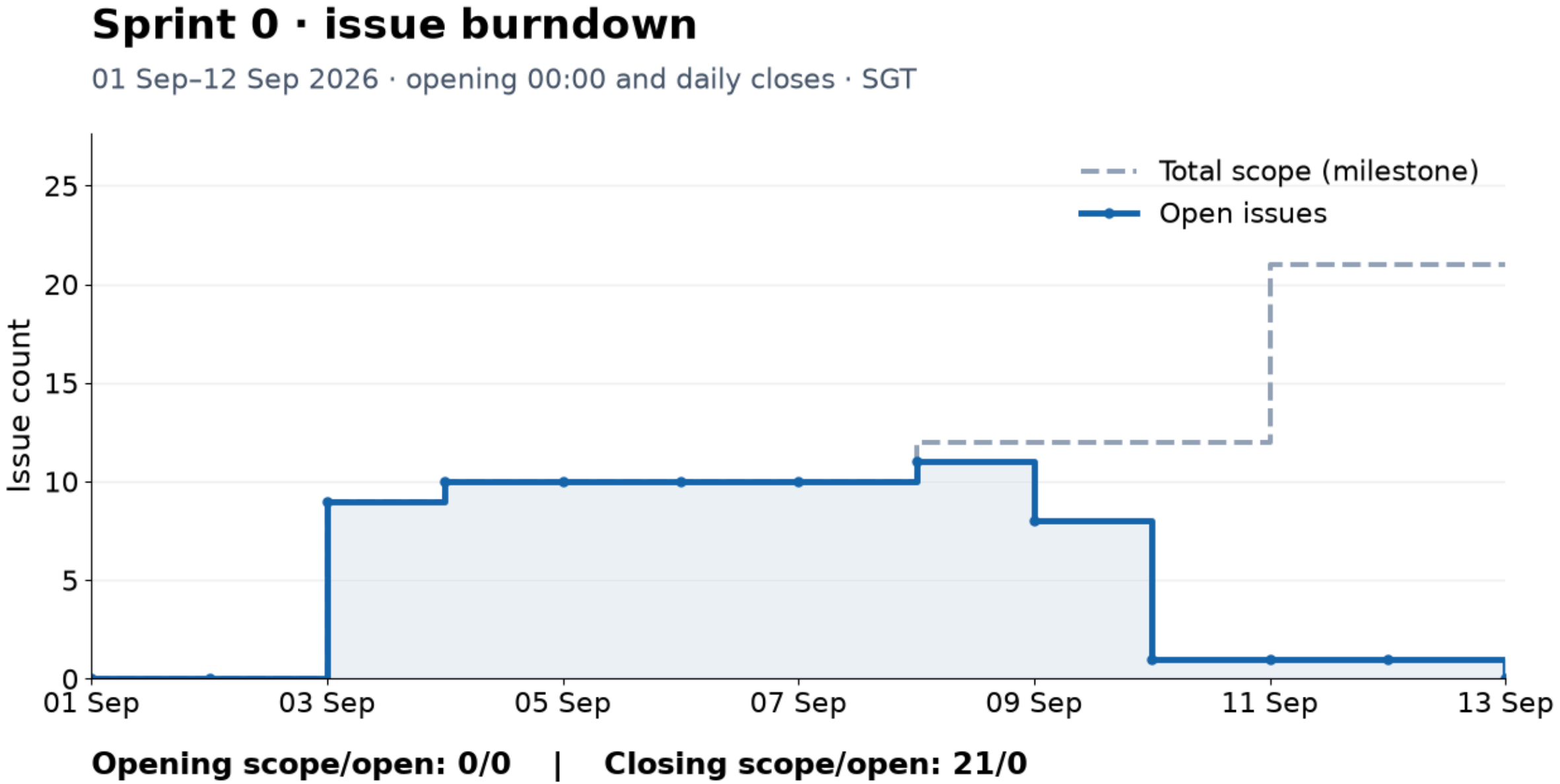
Use case model · UC04-UC06 Contribute within allowance



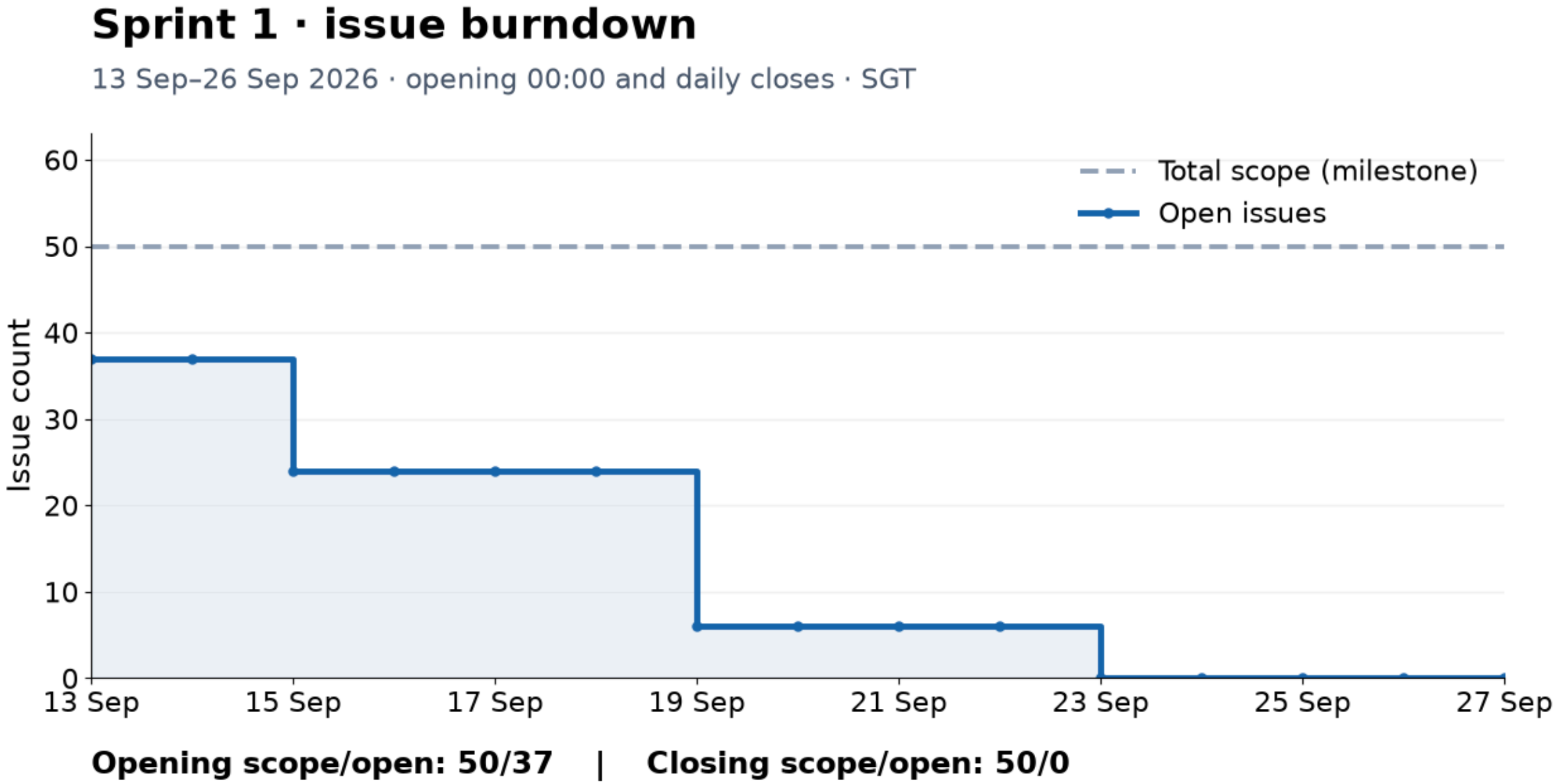
Use case model · UC07-UC09, UC12-UC13 Release and revisit the film



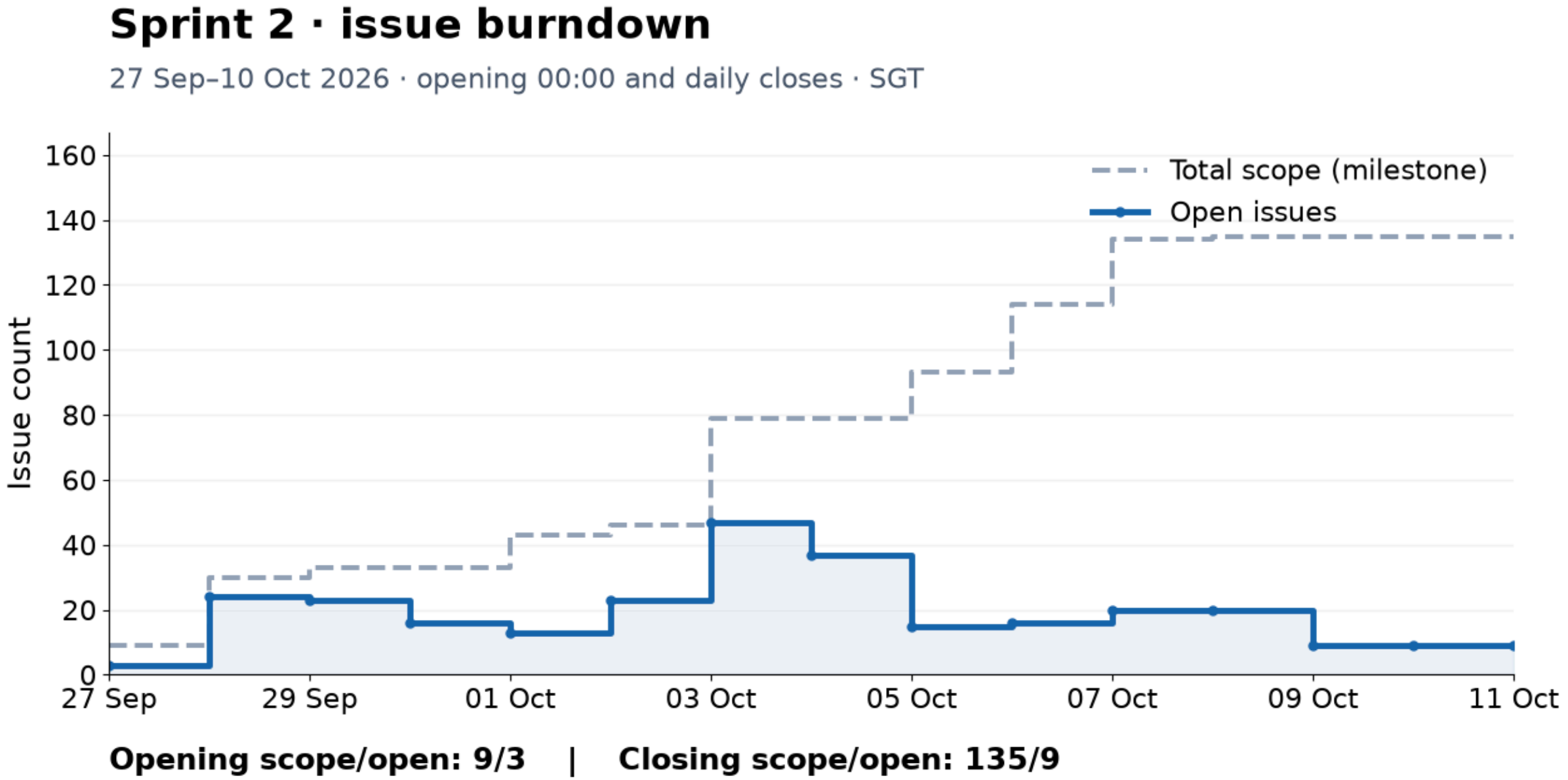
Sprint 0 records opening scope and daily delivery



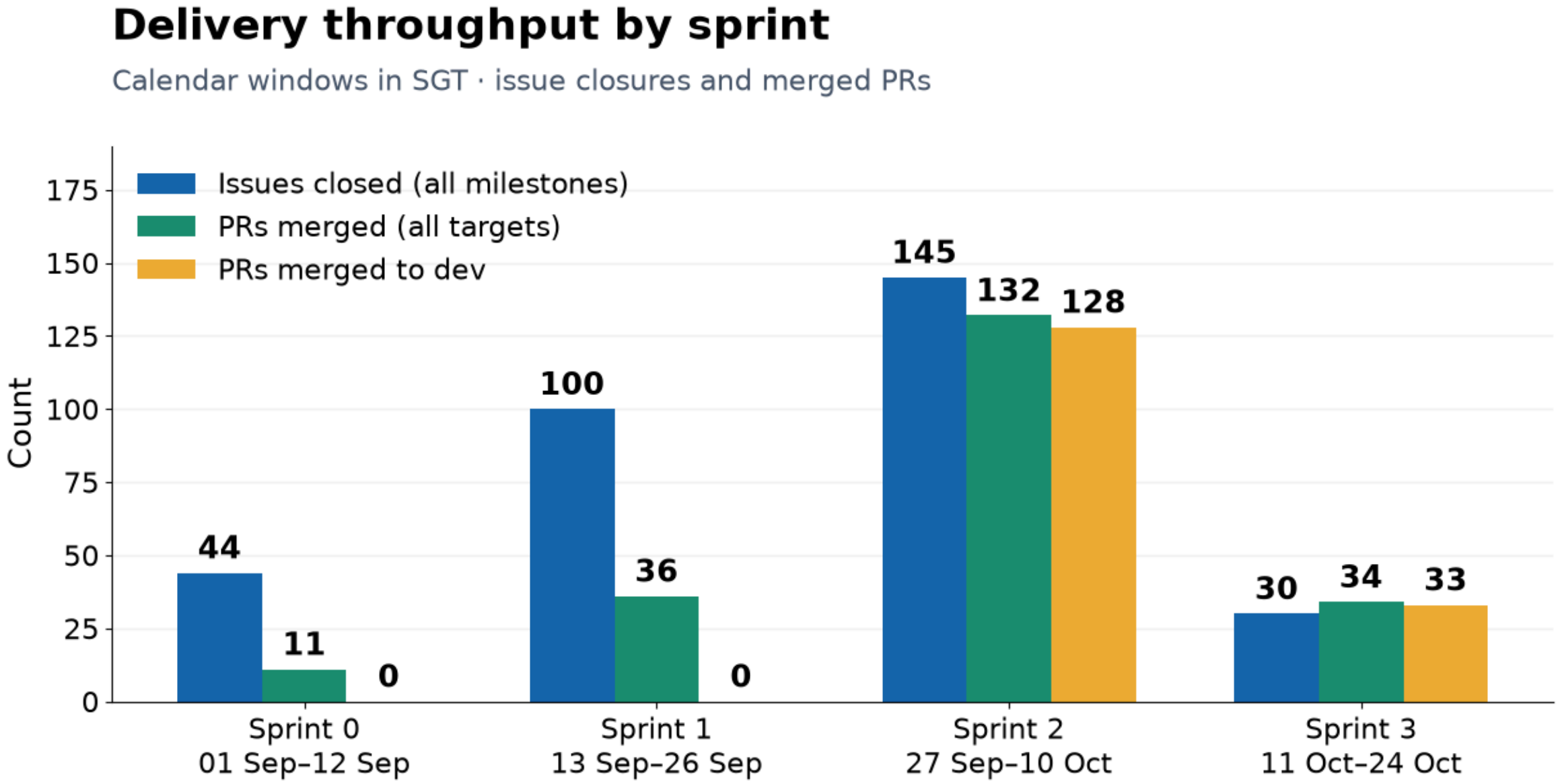
Sprint 1 records opening scope and daily delivery



Sprint 2 records opening scope and daily delivery



Issue closures and merged PRs track sprint delivery



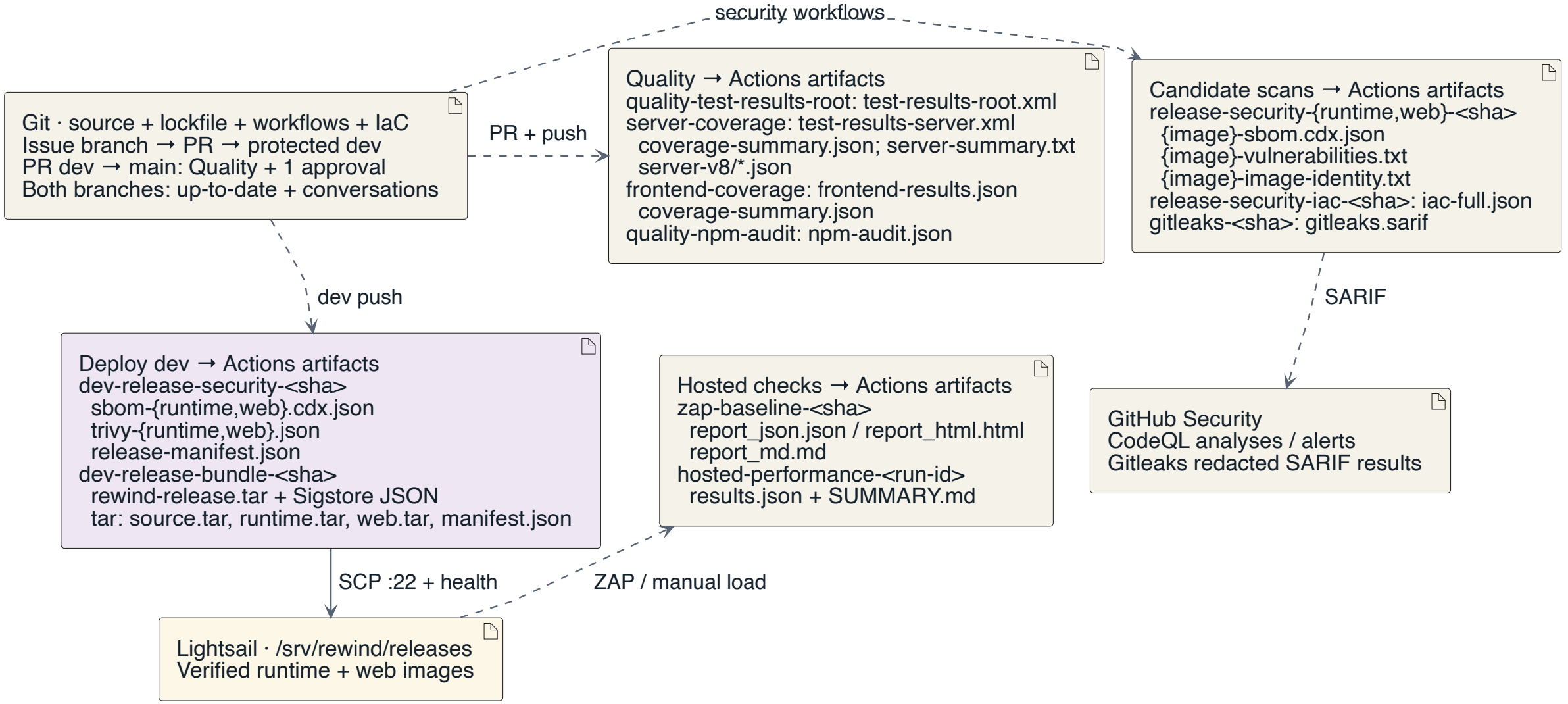
Backlog stories connect sprint goals to delivered use cases

Artifact	26 October record / trace
Product backlog	329 issues · 319 closed · 10 open; mvp delivery path has 7 open
User story · UC05	As a member, seal a photo as a three-second segment. S2 #313 → S3 #545 / #574
Sprint plans	S0 foundation → S1 integrated journey → S2 real accounts → S3 stability
Sprint Review	Each sprint reviews its increment against the selected scope
Retrospective	S1 integration gaps → S2 real adapters; S2 stale state → S3 interaction tests

Source: records/AGILE.md · milestones, story map and sprint plans

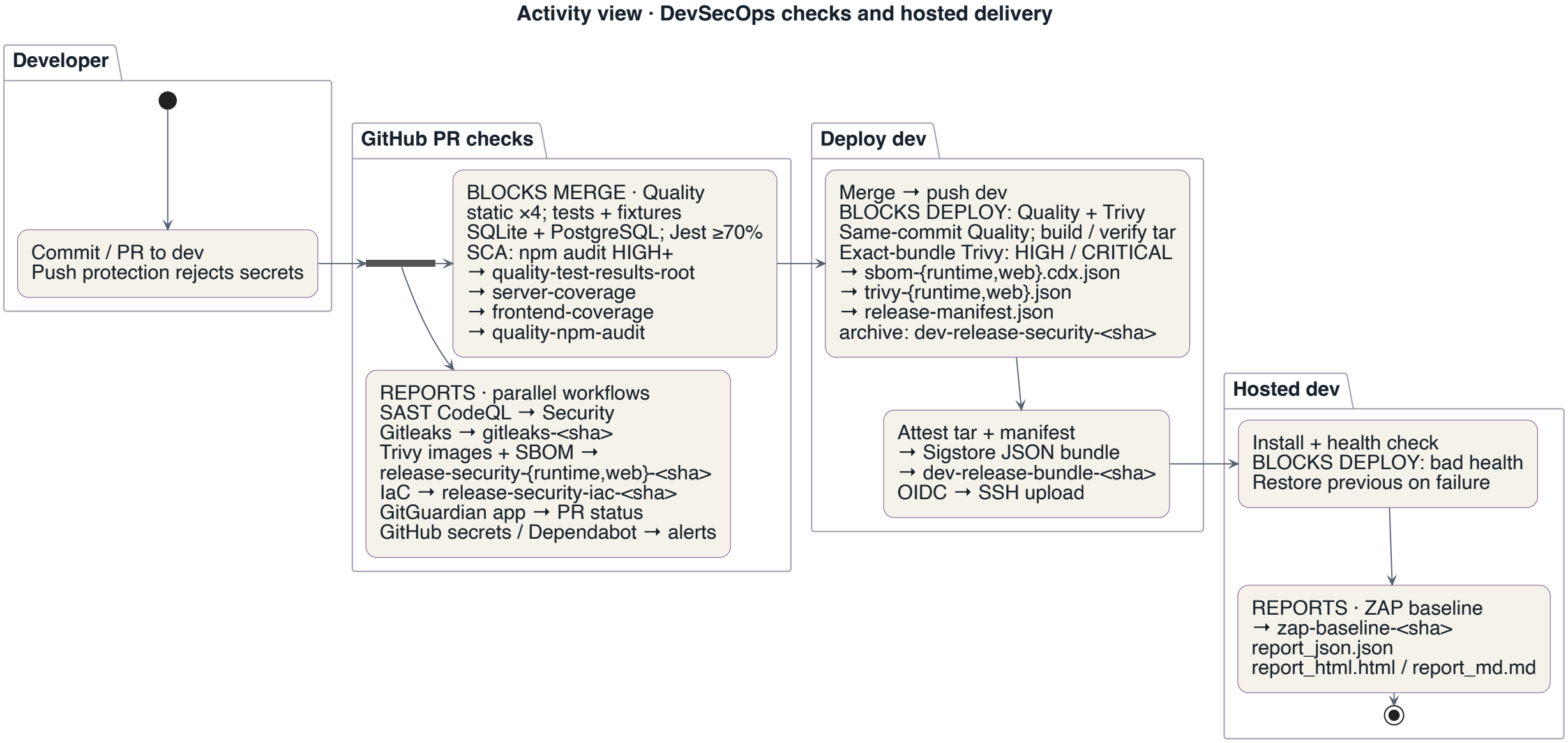
A commit carries tests, inventory and deployment evidence

Artifact view · Source, evidence and deployable bundle



Actions retention: 30 d; server/frontend coverage and hosted performance: 14 d. CodeQL results live in Security. Manifest binds commit + Quality run + image IDs + SHA-256(tar, SBOMs, Trivy). Attestation signs tar + manifest. <sha> = commit; {runtime,web} = both images. Main browser checks join Quality; hosted dev is the deployment target.

Pipeline decisions separate merge gates from host activation



BLOCKS MERGE = required Quality. BLOCKS DEPLOY = stops upload / activation. REPORTS = outside both gates.
PR + push: Quality, CodeQL, Gitleaks; CodeQL also weekly. Candidate scans: path-filtered PR, dev/main push, v*, manual.
Deploy: dev push / manual; docs-only skips host work and app checks. Main adds browser checks to Quality.
Manual performance REPORTS → hosted-performance-<run-id>: results.json + SUMMARY.md. Independent scans can fail their jobs.

Security checks cover code, packages, secrets and running pages

Procedure	Tool / scope	Result or policy	Decision
SAST	CodeQL · JS/TS, Python, Actions	3 load-script alerts → “used in tests”	Reports / triage
DAST	ZAP baseline · hosted dev	0 High · 4 Medium CSP · 7 Low · 6 Info	Post-deploy report
SCA	npm audit + Dependabot	Production audit HIGH+ joins Quality	Blocks merge
Secrets	Gitleaks + push protection	Full history scan + push-time secret checks	Scan + prevention
IaC	Trivy config · Terraform / Docker	15 findings · exact exceptions and expiry	Scan workflow gate
Images	Trivy · exact bundled runtime + web	0 High / Critical on both images	Blocks host upload

ZAP run 40344230832 · the four Medium findings concern Content Security Policy.

The manifest binds deployed images to their scans

Real excerpt · release-manifest.json · run 40344230832


```
{
  "commit": "aea7478bce14131b821caaa99b1185077b4a9403",
  "run_id": "40344230832",
  "quality_run_id": "40344231052",
  "security_gate": "pass",
  "sha256": {
    "rewind-release.tar":
    "fef3c22a19a2e5ef8f64cba8b074dc1fc2f3f3bd9ff102831e7ea387a7696e01"
  }
}
```


Runtime 0 H / 0 C · 1 M · 2 Unknown

Web 0 H / 0 C · 0 M

Inventory. CycloneDX SBOM for each bundled image.

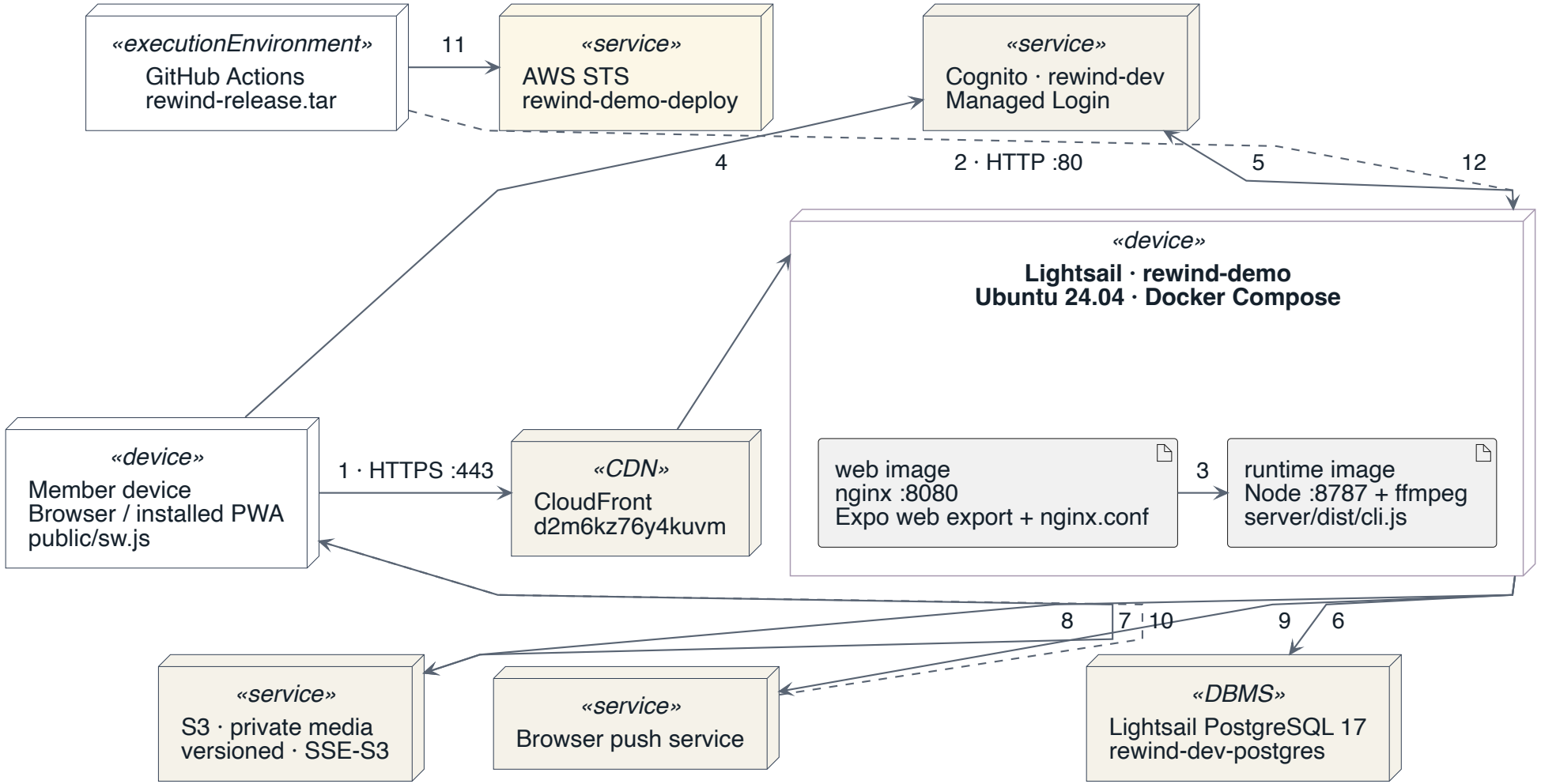
Identity. Commit, Quality run, image IDs, archive digests and report hashes.

Provenance. GitHub-signed attestations bind the bundle and manifest to the workflow.

Verify. Use gh attestation verify on the downloaded manifest.

Hosted dev activates after candidate health succeeds

Deployment view · Hosted dev



1 Browser → CloudFront: HTTPS :443. 2 CloudFront → nginx: HTTP :80 → :8080 + origin auth header.
3 nginx → Node: HTTP :8787 /api + SSE. 4 Browser → Cognito: HTTPS :443, authorization code + PKCE.
5 Node → Cognito: HTTPS :443 token exchange. 6 Node → PostgreSQL: TLS :5432.
7 Browser → S3: HTTPS :443 presigned PUT. 8 Node → S3: HTTPS :443 with IAM user credentials.
9 Node → push service: HTTPS :443 / VAPID. 10 Push service → PWA: push event.
11 Actions → STS: OIDC / HTTPS :443, 1 h dev role. 12 Actions → host: SCP / SSH :22, runner /32.
AWS ap-southeast-1. OIDC: dev branch + environment. Node owns the cycle, media and reminder loops.

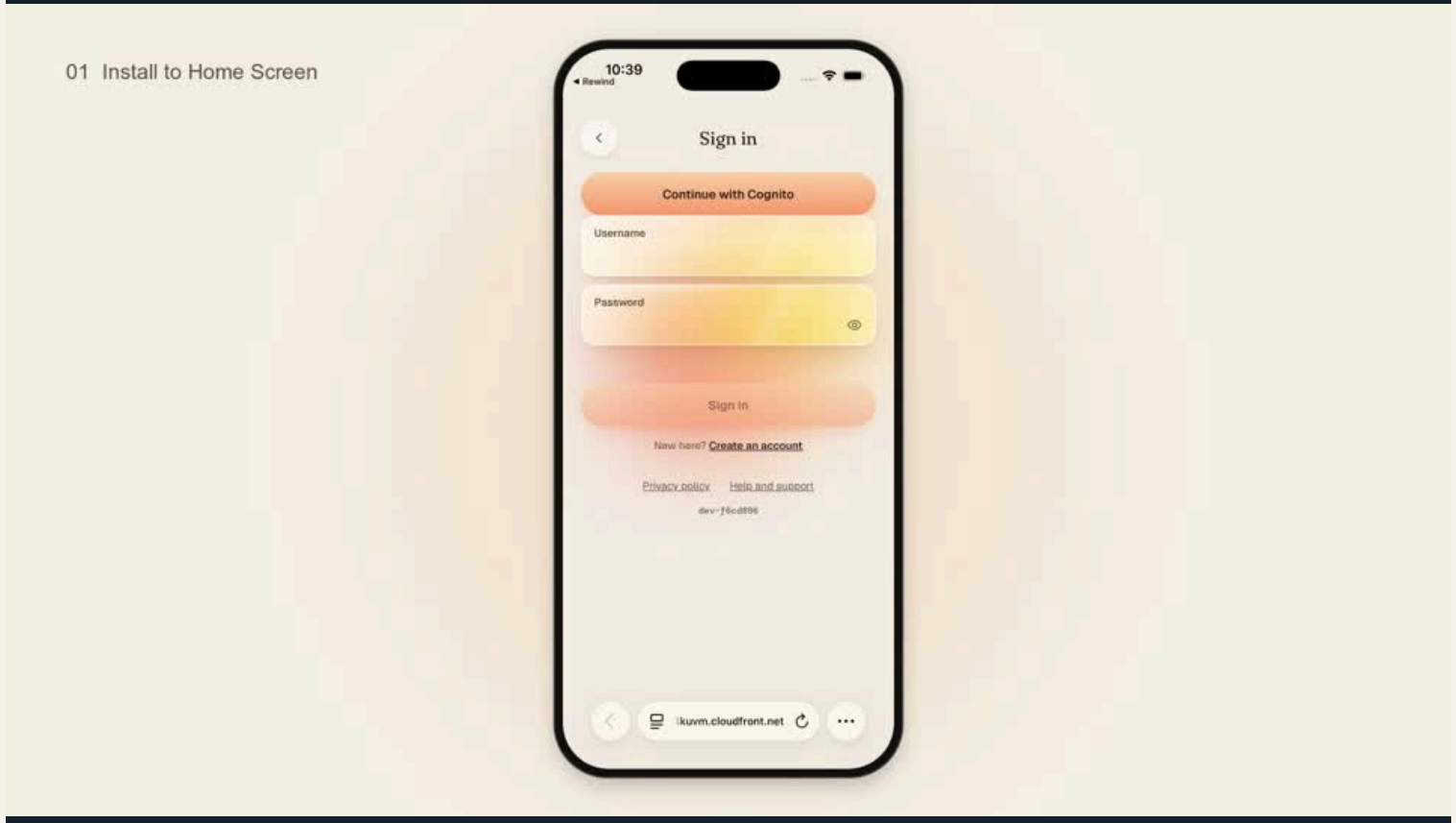
Packaging and runtime boundaries exposed useful defects

Finding	Consequence	Repair / learning
New advisory in a locked dependency tree	3 High zod findings blocked Quality	Override update + clean audit
Alpine removed a pinned FFmpeg package	Image build failed before scan	Update to available 8.1.3-r0 pin
Schema registry moved during refactoring	Bundle verifier read the old location	Read the declared schema version
Service worker retained SSE forwarding	Group switching exhausted streams	Browser-owned SSE + cancellation checks
Retry UI preserved the failed state	“Try again” returned without capture	Behavioural test + viewfinder reset

Test the adapter, the built image and the whole interaction.

DEMO · SECTION 5

Installation keeps the app one tap away



Do. Safari → Share → Add to Home Screen → open.

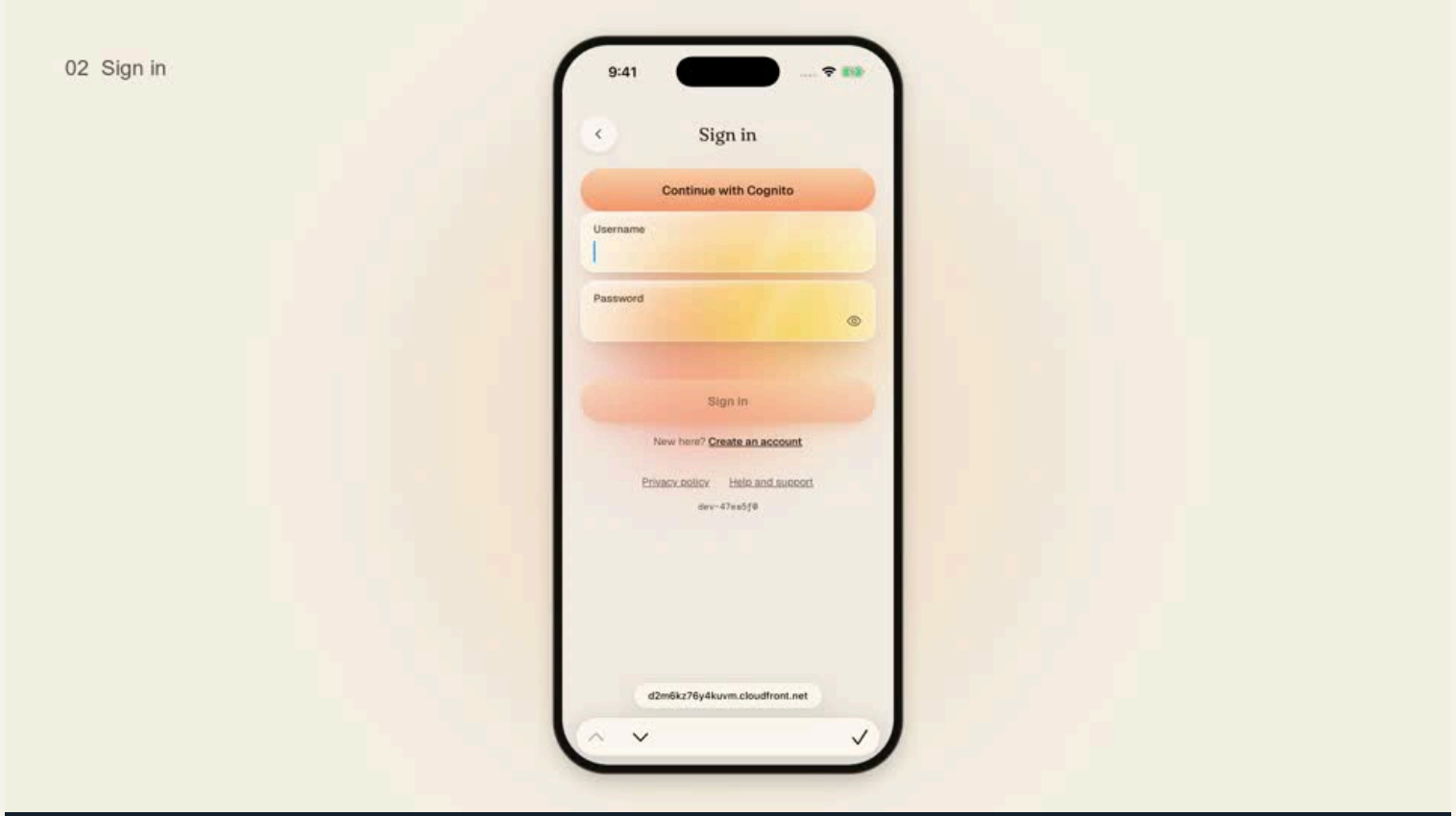
Observe. The installed app opens in its own Home Screen window.

Why. This is the iPhone delivery surface.

P / click · play or pause
26 OCT 2026 · HOSTED DEV

DEMO · SECTION 5

Sign-in returns the member to the private-group journey



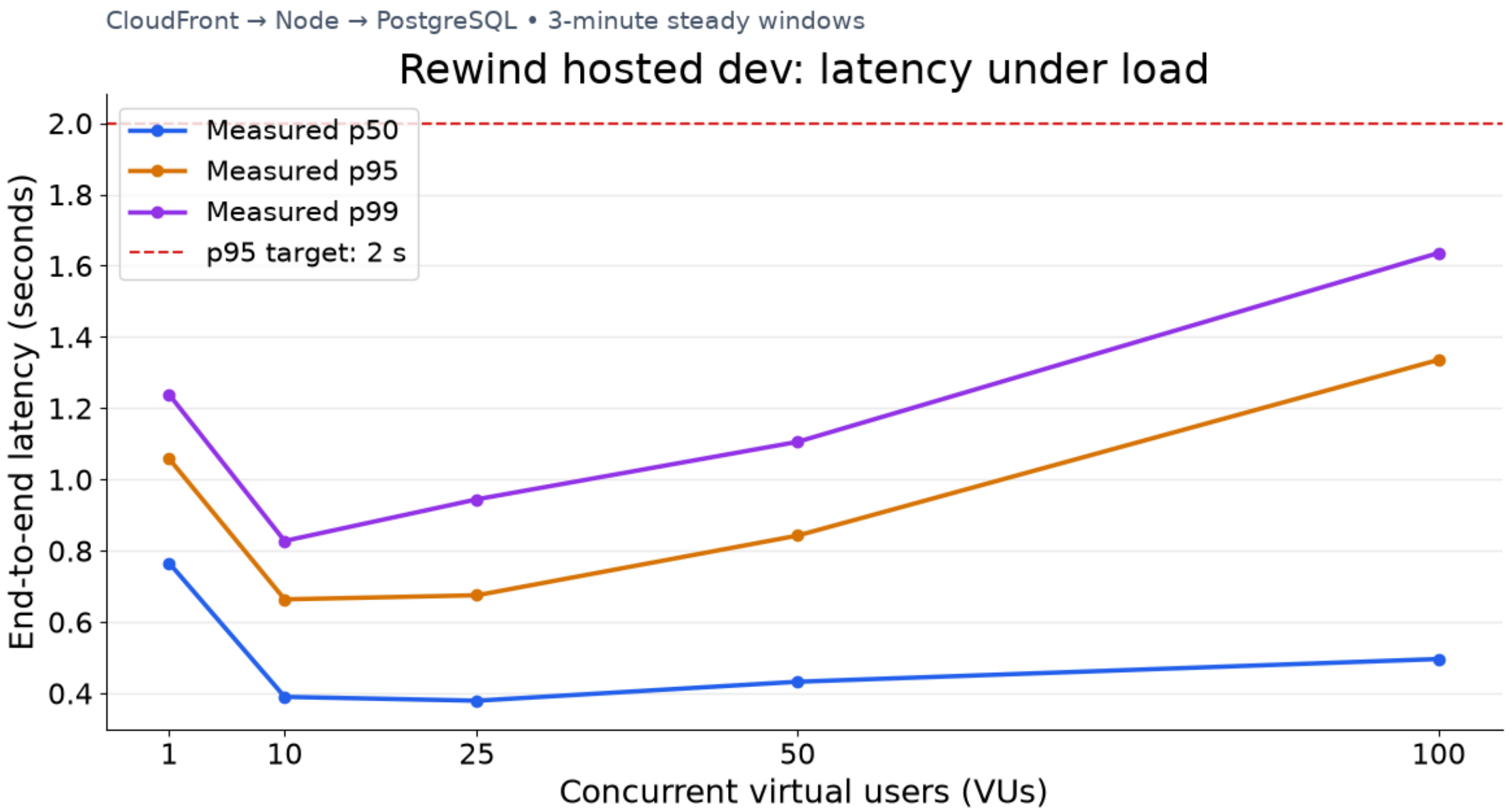
Do. Sign in as Maya and open her group space.

Observe. The authenticated account returns to Rewind.

Design. Web identity supports Cognito code + PKCE with server-side exchange.

P / click · play or pause
26 OCT 2026 · HOSTED DEV

Every steady load stage stays below the latency target



Measured throughput rises with the active user count

